Journal of Scientific Computing (2026) 106:2
https://doi.org/10.1007/s10915-025-03120-8

Efficient High-Order Explicit Local Time-Stepping Schemes for Dual-Horizon Peridynamics

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Received: 3 September 2024 / Revised: 8 July 2025 / Accepted: 2 November 2025
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Abstract

Peridynamics (PD) theory has shown great promise in fracture simulation due to its ability to accommodate discontinuities in the displacement field. Due to its simplicity and computational efficiency, explicit time-stepping is a popular way for simulating unsteady-state peridynamic (PD) problems, particularly those involving extensive material failure. Unlike conventional partial differential equation problems, the time step size in PD models is only restricted by the horizon size rather than the spatial grid size, owing to the special form of its Courant-Friedrichs-Lewy (CFL) condition. For dual-horizon PD models, where the horizon varies to deal with multiscale phenomenon with coarse and fine resolved regions, global time-stepping schemes could be very efficient when material points possess highly different horizon sizes. In this paper, we propose a family of high-order explicit local time-stepping (LTS) schemes for dual-horizon bond-based and ordinary stated-based PD based on the Runge-Kutta method. This approach allows for fine time steps in regions of interest to enhance accuracy, while the remainder of the domain can be solved with coarser time steps. We provide a rigorous convergence analysis for the proposed LTS schemes and perform numerical experiments to verify the expected convergence orders. Additionally, the PD simulation results, including those of plates with pre-existing cracks and the 3D Kalthoff-

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Published online: 02 December 2025

Winkler experiment, demonstrate effectiveness of the proposed method in achieving accurate fracture simulations.

Keywords Peridynamics · Local time-stepping · Dual-horizon peridynamics · Crack propagation

1 Introduction

Classical continuum mechanics, grounded in partial differential equations, faces significant hurdles when modeling systems characterized by discontinuities. In response, peridynamics (PD) is an integral nonlocal model that addresses these discontinuity challenges comprehensively. Over recent decades, PD has spurred extensive research in modeling, digital technologies, and practical applications. These applications range from composite material deformation [1–4] and addressing corrosion [5–8], to predicting damage [9–13] and simulating cracks in various materials [14–18]. The initial PD models are fixed horizon models, including bond-based PD [19, 20], ordinary state-based PD [21], and non-ordinary state-based PD [22]. However, spurious wave reflections and ghost forces would arise when using these PD models with an irregular particle distribution, decreasing numerical precision. To address this issue, variable horizon peridynamics, also known as dual-horizon peridynamics (DH-PD), was first proposed in [23]. This approach improves numerical accuracy and minimizes the number of discrete particles in fracture mechanics problems. It should be noted that horizon is guided by the characteristic range of long-range forces [19, 20], not the discretization of the computational domain.

For simulating dynamic peridynamic (PD) problems, explicit time-stepping methods such as the velocity Verlet method [24] and the Runge-Kutta method [25] are popular choices for temporal discretization. Adherence to a CFL condition is crucial to ensure numerical stability in these methods. According to [24], the horizon size determines the CFL condition in PD. For instance, in one-dimensional bond-based PD, the CFL condition can be expressed as:

$$\Delta t < \sqrt{\frac{2\rho}{\sum_p V_p C_{ip}}}, \tag{1}$$

where V_p represents the volume of material point $\mathbf{x}_p$, and C_{ip} is determined by the micromodulus function. Therefore, in the dual-horizon peridynamics (DH-PD) model, the time step size is influenced by the minimum horizon size, which can result in significant computational costs for multiscale problems. To tackle this challenge, we introduce explicit local time-stepping (LTS) algorithms in this work. These algorithms enable the time step size to adapt to locally varying CFL conditions rather than being constrained by a global CFL condition.

The initial LTS method is firstly introduced in [26] for one-dimensional scalar conservation laws based on the forward Euler discretization and its efficacy have been confirmed when integrated with local spatial refinement in 2D and 3D scenarios [27–29]. Following that, research has delved into exploring the LTS method within the framework of discontinuous Galerkin discretization for the shallow water equations [30, 31], and the LTS method combined with the finite volume method is examined in [32, 33]. Additionally, explicit LTS methods based on high-order Runge-Kutta (RK) techniques were proposed for conservation laws in [34] and for wave propagation in [35]. A high-order LTS method is proposed for the baroclinic-barotropic split dynamics in primitive equations [36].In [37], the LTS method

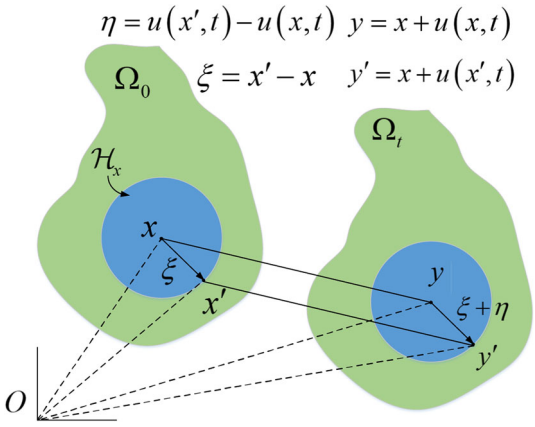


Fig. 1 The configuration for deformed body

was first applied to bond-based PD. Based on the Newmark-β, this method partitions the domain into discrete overlapping subdomains and employs different time step sizes during time discretization. Subsequently, [38] proposed an higher orders LTS method for PD using the fourth-order Runge-Kutta method, achieving higher accuracy with less computational cost. However, there is still an absence of the LTS method for DH-PD.

The proposed LTS method is designed for bond-based and ordinary stated-based DH-PD models with varying horizon sizes and encompasses second-order, third-order, and fourth-order schemes. To construct the LTS method, we partition the Cartesian region into regions of interest and others, employing different time step sizes based on the horizon size. Regions of interest utilize fine time steps to ensure calculation accuracy, while other regions employ coarse time steps to enhance computational efficiency. To handle the coupling between coarse and fine time steps and to approximate interface values at intermediate time levels, we introduce an interpolation function derived from Taylor series expansions of the solution at the current time level. Convergence analysis confirms that our second-order, third-order, and fourth-order LTS methods maintain the convergence order of the original explicit Runge-Kutta (RK) method. Subsequently, we validate the performance of our proposed methods on discontinuous problems through experiments on crack propagation in a board and a three-dimensional Kalthoff-Winkler experiment.

The rest of the article is organized as follows: The second section reviews the fundamental theories of PD. Following that, the third section introduces the DH-PD. Within the third section, a theoretical derivation and analysis of the LTS method applicable to the DH-PD are presented. The fourth section validates the efficiency and accuracy of the proposed algorithm through numerical examples.

2 Fundamentals of Peridynamics and its Discretization

The PD is a dynamic approach based on non-local concepts, asserting that a material point is active within a finite influence region $\mathcal{H}_x$ determined by its action range δ, as illustrated in Fig. 1. Here, $\mathbf{y}$ and $\mathbf{y}'$ represent the positions of the transformed material points $\mathbf{x}$ and $\mathbf{x}'$, respectively.

The motion equation for PD [19, 20] can be defined as follows

$$\rho\ddot{\mathbf{u}}(\mathbf{x},t)=\int_{\mathcal{H}_{\mathbf{x}}}\mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi})\mathrm{d}V_{\mathbf{x}'}+\mathbf{b}(\mathbf{x},t), \tag{2}$$

where ρ represents the mass density, $\mathbf{b}(\mathbf{x},t)$ denotes the body force density field, and $\mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi})$ is a pairwise force function used to compute the force vector. Notably, $\mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi})$ satisfies the condition $\mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi})=-\mathbf{f}(-\boldsymbol{\eta},-\boldsymbol{\xi})$. $\mathcal{H}_{\mathbf{x}}$ is usually taken as a disk in 2D or ball in 3D and can be defined by

$$\mathcal{H}_{\mathbf{x}}=\left\{\mathbf{x}'\mid\left\|\mathbf{x}'-\mathbf{x}\right\|<\delta_{\mathbf{x}}\right\}, \tag{3}$$

here $\delta_{\mathbf{x}}$ is the radius of $\mathcal{H}_{\mathbf{x}}$.

The primary peridynamic equations encompass two types: bond-based peridynamics (BB-PD) and state-based peridynamics (SB-PD). In this paper, we focus on BB-PD and ordinary state-based peridynamics (OSB-PD). For BB-PD, the material points interact with each other within the $\mathcal{H}_{\mathbf{x}}$, where the magnitude and direction of the force density vector parallel to the relative position vector in the deformation state are identical and can be expressed as follows:

$$\mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi})=\mu\alpha s(\boldsymbol{\eta},\boldsymbol{\xi})\cdot\frac{\boldsymbol{\eta}+\boldsymbol{\xi}}{\|\boldsymbol{\eta}+\boldsymbol{\xi}\|},\quad \alpha=\begin{cases}\frac{9E}{\pi\delta^3 h}, & \text{2D},\\ \frac{12E}{\pi\delta^4}, & \text{3D},\end{cases}\quad s(\boldsymbol{\eta},\boldsymbol{\xi})=\frac{\|\boldsymbol{\eta}+\boldsymbol{\xi}\|-\|\boldsymbol{\xi}\|}{\|\boldsymbol{\xi}\|}. \tag{4}$$

here μ is a history-dependent scalar-valued function, s_0 is the the critical bond stretch, and

$$\mu=\begin{cases}1, & s(\boldsymbol{\eta},\boldsymbol{\xi})<s_0 \text{ for all } 0\le t'\le t,\\ 0, & \text{otherwise},\end{cases},\quad s_0=\begin{cases}\sqrt{\frac{4\pi G_0}{9E\delta}}, & \text{plane stress},\\ \sqrt{\frac{5\pi G_0}{12E\delta}}, & \text{plane strain},\end{cases} \tag{5}$$

where G_0 is the energy release rate, h is the plate thickness for the 2D model.

For OSB-PD, the force density vectors have unequal magnitudes while parallel to the relative position vector in the deformed state. Thus, a dilation of material point $\mathbf{x}$, denoted as $\theta(\mathbf{x},t)$, is introduced to describe the force density function, where the $\mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi})$ can be defined by the following

$$\mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi})=\mu\left(\beta\frac{\boldsymbol{\eta}+\boldsymbol{\xi}}{\|\boldsymbol{\eta}+\boldsymbol{\xi}\|}\cdot\frac{\boldsymbol{\xi}}{\|\boldsymbol{\xi}\|^2}\left(\theta(\mathbf{x},t)+\theta\left(\mathbf{x}',t\right)\right)+\gamma\frac{\|\boldsymbol{\xi}+\boldsymbol{\eta}\|-\|\boldsymbol{\xi}\|}{\|\boldsymbol{\xi}\|}\right)\frac{\boldsymbol{\xi}+\boldsymbol{\eta}}{\|\boldsymbol{\xi}+\boldsymbol{\eta}\|}, \tag{6}$$

with

$$\beta=\frac{3K-5G}{m},\quad \gamma=-\frac{30G}{m},\quad \theta(\mathbf{x}',t)=\frac{3}{m}\int_{\mathcal{H}_{\mathbf{x}}}\omega(\|\boldsymbol{\xi}\|)\boldsymbol{\xi}\cdot\boldsymbol{\eta}\mathrm{d}V_{x'}, \tag{7}$$

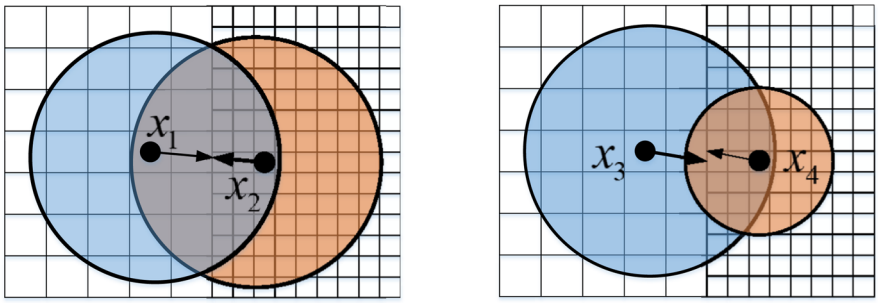
with G being the shear modulus, K being the bulk modulus, and

$$m=\int_{\mathcal{H}_{\mathbf{x}}}\omega(\|\boldsymbol{\xi}\|)\|\boldsymbol{\xi}\|^2\,\mathrm{d}V_{\mathbf{x}'},\quad \omega(\|\boldsymbol{\xi}\|)=\frac{1}{\|\boldsymbol{\xi}\|}. \tag{8}$$

Let $\mathbf{x}_j$ and $\mathbf{x}_j$ denote two distinguish discrete material points of the motion equation, and $\mathbf{u}_j$ and $\mathbf{u}_i$ represent their respective displacements. To discretize $\mathbf{f}$ at a given material point $\mathbf{x}_p$, a meshfree method [24] is employed due to its simplicity as follows:

$$\rho\ddot{\mathbf{u}}_i=\sum_{j=1}^{K}\mathbf{f}(\mathbf{u}(\mathbf{x}_j,t)-\mathbf{u}(\mathbf{x}_i,t),\mathbf{x}_j-\mathbf{x}_i)V_{\mathbf{x}_j}, \tag{9}$$

where K represents the total number of $\mathbf{x}_j$ within the horizon $\mathcal{H}_{\mathbf{x}_i}$, and $V_{\mathbf{x}_j}$ represents the volume of $\mathbf{x}_j$.



(a) Geometry with constant horizon (b) Geometry with variable horizon

Fig. 2 Illustration of constant horizon and variable horizon with $r_{\mathbf{x}}$ denote the mesh spacing and $V_{\mathbf{x}} = r_{\mathbf{x}}^2$: (a) The neighbors of material points $\mathbf{x}_1$ and $\mathbf{x}_2$ with constant horizon $\delta_{\mathbf{x}_1} = \delta_{\mathbf{x}_2} = 3r_{\mathbf{x}_1} = 6r_{\mathbf{x}_2}$; (b)The neighbors of material points $\mathbf{x}_i$ and $\mathbf{x}_j$ with variable horizon $\delta_{\mathbf{x}_3} = 2\delta_{\mathbf{x}_4} = 3r_{\mathbf{x}_3} = 6r_{\mathbf{x}_4}$

3 Dual-Horizon Peridynamics

In traditional PD, the horizon is chosen to be constant. However, in the realm of multi-scale problems, this fixed horizon size δ introduces computational costs and accuracy issues. Therefore, instead of a constant horizon size δ, a variable horizon size $\delta_{\mathbf{x}_i}$ is tend to be chosen for for different material points.

3.1 The Shortcoming of Constant Horizon and Ghost Force

One fundamental limitation of traditional PD is the requirement for constant radii of horizons, resulting in high computational costs. Moreover, the use of variable horizons can adversely affect accuracy.

In the case of constant horizon, as depicted in Fig. 2 (a), the discretization of the computational domain with varying material point sizes results in a significant increase in the number of neighbors within the horizon of $\mathbf{x}_2$ compared to that of $\mathbf{x}_1$. This not only escalates computational costs but also minimally contributes to accuracy enhancement, as the values at each point are smoothed out by their numerous neighbors.

With variable horizons, as illustrated in Fig. 2 (b), an imbalance arises in the interaction forces between material points $\mathbf{x}_3$ and $\mathbf{x}_4$, where forces from $\mathbf{x}_3$ to $\mathbf{x}_4$ occur, but the reverse interaction from $\mathbf{x}_4$ to $\mathbf{x}_3$ cannot be established. This asymmetry results in the emergence of ghost forces, where the applied force lacks an equal reaction force. Consequently, the foundational assumption of the PD model regarding the pairing of forces between bonded points is compromised. Hence, there is a pressing need for a novel model to address the challenges posed by variable horizons.

3.2 Horizon and Dual-Horizon

In order to deal with ghost forces, a DH-PD was introduced. In the DH-PD, alongside the primary horizon, an additional dual horizon is introduced. For a given material point $\mathbf{x}$, the dual horizon can be represented as the collective set of material points within its primary

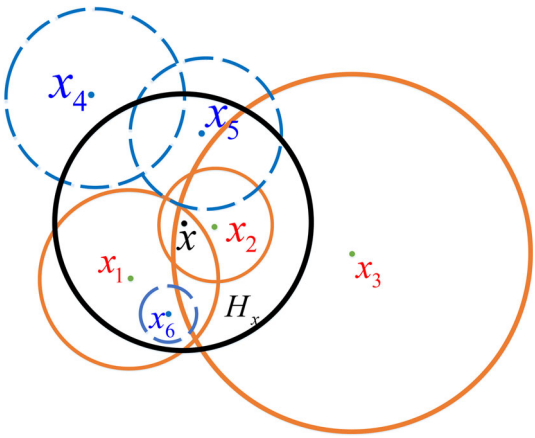


Fig. 3 The schematic diagram illustrates the concept of both the primary horizon and dual-horizon. The primary horizon of material point $\mathbf{x}$ is denoted by a black line. The red points $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\} \in \mathcal{H}'_{\mathbf{x}}$ since $\mathbf{x}$ lie in their primary horizon denoted by the red solid lines. Similarly, The red points $\{\mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6\} \notin \mathcal{H}'_{\mathbf{x}}$ since $\mathbf{x}$ doesn't lie in their primary horizon denoted by the blue dotted lines

horizon that encompass it, denoted as:

$$\mathcal{H}'_{\mathbf{x}} = \left\{\mathbf{x}' \mid \left\|\mathbf{x}' - \mathbf{x}\right\| < \delta_{\mathbf{x}'}\right\}, \tag{10}$$

which means $\mathcal{H}'_{\mathbf{x}} = \left\{\mathbf{x}' \mid \mathbf{x} \in \mathcal{H}_{\mathbf{x}'}\right\}$. Here, the superscript prime represents 'dual', and the subscript represents the central material point. The shape of the dual horizon is arbitrary, depending on both the primary horizon size and the position of the material points around $\mathbf{x}$, and is usually not a disk or ball, as shown in Fig. 3.

By introduce the dual-horizon, we can rewritten the reaction force $\mathbf{f}(-\boldsymbol{\eta}, -\boldsymbol{\xi})$ and change the equation of motion as

$$\rho\ddot{\mathbf{u}}(\mathbf{x}, t) = \frac{1}{2}\int_{\mathcal{H}_{\mathbf{x}}} \mathbf{f}(\boldsymbol{\eta}, \boldsymbol{\xi})\, \mathrm{d}V_{\mathbf{x}'} + \frac{1}{2}\int_{\mathcal{H}'_{\mathbf{x}}} \mathbf{f}(-\boldsymbol{\eta}, -\boldsymbol{\xi})\, \mathrm{d}V_{\mathbf{x}'} + \mathbf{b}(\mathbf{x}, t), \tag{11}$$

For BB-PD, the direct force and reaction force can be expressed as

$$\begin{aligned} \mathbf{f}(\boldsymbol{\eta}, \boldsymbol{\xi}) &= \mu\alpha\left(\delta_{\mathbf{x}}\right) \cdot s(\boldsymbol{\eta}, \boldsymbol{\xi}) \cdot \frac{\boldsymbol{\eta} + \boldsymbol{\xi}}{\|\boldsymbol{\eta} + \boldsymbol{\xi}\|}, \quad \forall \mathbf{x}' \in H_{\mathbf{x}} \\ \mathbf{f}(-\boldsymbol{\eta}, -\boldsymbol{\xi}) &= \mu\alpha\left(\delta_{\mathbf{x}'}\right) \cdot s(-\boldsymbol{\eta}, -\boldsymbol{\xi}) \cdot \frac{-(\boldsymbol{\eta} + \boldsymbol{\xi})}{\|\boldsymbol{\eta} + \boldsymbol{\xi}\|}, \quad \forall \mathbf{x}' \in H'_{\mathbf{x}}, \end{aligned} \tag{12}$$

with

$$\alpha\left(\delta_{\mathbf{x}}\right) = \begin{cases} \dfrac{9E}{\pi\delta_{\mathbf{x}}^3 h}, & \text{2D}, \\ \dfrac{12E}{\pi\delta_{\mathbf{x}}^4}, & \text{3D}, \end{cases} \tag{13}$$

For OSB-PD, the reaction force can be expressed as

$$\begin{aligned}
\mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi}) &= \mu\left(\beta(\mathbf{x})\frac{\boldsymbol{\eta}}{\|\boldsymbol{\eta}\|}\cdot\frac{\boldsymbol{\xi}}{\|\boldsymbol{\xi}\|^2}\left(\theta(\mathbf{x},t)+\theta\left(\mathrm{x}',t\right)\right)+\gamma(\mathbf{x})\frac{\|\boldsymbol{\xi}+\boldsymbol{\eta}\|-\|\boldsymbol{\xi}\|}{\|\boldsymbol{\xi}\|}\right)\\
&\quad\frac{\boldsymbol{\xi}+\boldsymbol{\eta}}{\|\boldsymbol{\xi}+\boldsymbol{\eta}\|},\quad \forall\mathbf{x}'\in H_{\mathbf{x}}\\
\mathbf{f}(-\boldsymbol{\eta},-\boldsymbol{\xi}) &= \mu\left(\beta(\mathbf{x}')\frac{\boldsymbol{\eta}}{\|\boldsymbol{\eta}\|}\cdot\frac{\boldsymbol{\xi}}{\|\boldsymbol{\xi}\|^2}\left(\theta(\mathbf{x},t)+\theta\left(\mathrm{x}',t\right)\right)+\gamma(\mathbf{x}')\frac{\|\boldsymbol{\xi}+\boldsymbol{\eta}\|-\|\boldsymbol{\xi}\|}{\|\boldsymbol{\xi}\|}\right)\\
&\quad\frac{\boldsymbol{\xi}+\boldsymbol{\eta}}{\|\boldsymbol{\xi}+\boldsymbol{\eta}\|},\quad \forall\mathbf{x}'\in H'_{\mathbf{x}},
\end{aligned}\tag{14}$$

where

$$\beta(\mathbf{x})=\frac{3K-5G}{m_{\mathbf{x}}},\quad \gamma(\mathbf{x})=-\frac{30G}{m_{\mathbf{x}}},\quad \theta(\mathbf{x}',t)=\frac{3}{m_{\mathbf{x}}}\int_{\mathcal{H}_{\mathbf{x}}}\omega(\|\boldsymbol{\xi}\|)\boldsymbol{\xi}\cdot\eta\mathrm{d}V_{x'},\tag{15}$$

and

$$m_{\mathbf{x}}=\int_{\mathcal{H}_{\mathbf{x}}}\omega(\|\boldsymbol{\xi}\|)\|\boldsymbol{\xi}\|^2\,\mathrm{d}V_{\boldsymbol{x}'}.\tag{16}$$

Remark 1 Unlike the classical continuum mechanics equations, where the CFL condition is determined by the mesh size, the CFL condition in the discretized peridynamic (PD) framework is governed by the horizon size. Here, we take the velocity-Verlet method as an example. According to [24], applying the velocity-Verlet scheme to the equivalent equation yields:

$$m\frac{y_j^{n+1}-2y_j^n+y_j^{n-1}}{(\Delta t)^2}=\sum_i f_{i,j},\tag{17}$$

with m the particle mass, Δt the time step, $y_j^n\equiv y\left(x_j,t^n\right)$ the position at time t^n of a particle that was at x_j in the reference configuration, and $f_{i,j}$ the force that particle i exerts on particle j. By substituting the force density function and letting $y_j^n=x_j+\zeta^n e^{ikj\Delta x}$, we have

$$\Delta t<\delta\sqrt{\frac{\rho}{2\alpha}}\sqrt{\frac{1+\frac{1}{N}}{\ln(N)+\gamma}}\approx\delta\sqrt{\frac{\rho}{2\alpha}}\sqrt{\frac{1}{\ln(N)}}\approx\delta\sqrt{\frac{\rho}{2\alpha}}\frac{1}{\sqrt{\ln\left(\frac{1}{\Delta x}\right)}},\tag{18}$$

where $\gamma>0$ is the Euler–Mascheroni constant. Here we assumed $\ln(1/\Delta x)\gg\ln(\delta)$ in the rightmost approximation. It is straightforward to show that $1/\sqrt{\ln(1/\Delta x)}>\Delta x$; therefore, stability condition of peridynamics implies that the dependence of Δt on Δx is weaker than the classical Courant-Friedrichs-Lewy (CFL) condition for the wave equation that requires $\Delta t=O(\Delta x)$ as $\Delta x\to 0$. On the other hand, in the limit $1/\Delta x\gg\delta$, stability condition of peridynamics reduces to $\Delta t=O(\delta)$. Therefore, when employing the DH-PD framework, the selection of the time step is influenced by the horizon sizes of different scales.

4 High-Order Explicit Local Time-Stepping Method for Dual-Horizon Peridynamics

The explicit algorithm, valued for its simplicity, finds common application in time integration for discrete systems. In this study, we focus on the explicit RK method, considering its second-, third-, and fourth-order accuracy for time discretization in the DH-PD. However, traditional

PD links the global CFL condition to the horizon size. This association constrains the time step based on the most miniature horizon in DH-PD at different resolutions, leading to an escalation in computational costs. To address this challenge, a LTS method is proposed. Relying on local CFL conditions allows distinct time steps in various spatial regions.

The following section introduces the global RK method with a single time step and outlines the LTS method, utilizing varying time steps in different regions.

4.1 Global Time-Stepping by the RK Methods

This section presents a high-order temporal discretization scheme for BB-PD. By employing a uniformly-sized partition of the time interval $[0, T]$ into N subintervals and denoting $\Delta t = T/N$, $t_i = i\Delta t$, we can use the RK method outlined in [25] to formulate a semi-discrete system corresponding to (9). For convenience, we will represent the semi-discrete system as follows:

$$\frac{d\mathbf{U}_i(t)}{dt} = \mathcal{L}_i\left(\mathbf{Q}(t)\right), \tag{19}$$

where $\mathbf{Q}(t) = \{\mathbf{U}_j(t), \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\}$, $\mathbf{U}_i(t) = \left[\mathbf{u}(\mathbf{x}_i, t), \mathbf{v}(\mathbf{x}_i, t)\right]^{\mathrm{T}}$, $\mathcal{L}$ denotes the spatial operator, defined as follows:

$$\mathcal{L}_i(\mathbf{Q}) = \begin{bmatrix} \mathbf{v}(\mathbf{x}_i, t) \\ \frac{1}{2\rho}\sum_{i}^{K}\mathbf{f}(\mathbf{u}(\mathbf{x}_j, t) - \mathbf{u}(\mathbf{x}_i, t), \mathbf{x}_j - \mathbf{x}_i)V_{\mathbf{x}_j} - \frac{1}{2\rho}\sum_{i}^{K'}\mathbf{f}(\mathbf{u}(\mathbf{x}_i, t) - \mathbf{u}(\mathbf{x}_j, t), \mathbf{x}_i - \mathbf{x}_j)V_{\mathbf{x}_j} \end{bmatrix}, \tag{20}$$

in which $\mathbf{v} = \dot{\mathbf{u}}$, K' represent the total number of the material points in $H'_{\mathbf{x}_i}$.

Let $\mathbf{U}_i^n = \mathbf{U}_i(t_n)$ and assume initial value $\mathbf{U}_i^n$, the *second order*, *two-stage* RK method, RK2 for solving (2) can be obtain by:
(i)Stage 1:

$$\mathbf{U}_i^{(1)} = \mathbf{U}_i^n + \Delta t\mathcal{L}_i\left(\mathbf{U}^n\right), \tag{21}$$

(ii) Stage 2:

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n + \frac{\Delta t}{2}\left(\mathcal{L}_i(\mathbf{U}_n) + \mathcal{L}_i\left(\mathbf{U}^{(1)}\right)\right). \tag{22}$$

Remark 2 When solving the PD, the RK2 method is equivalent to the velocity-Verlet (VV) method. We can provided a detailed proof below to illustrate this issue:

According to (21), the $\mathbf{U}_i^{(1)}$ can be expressed as

$$\mathbf{U}_i^{(1)} = \begin{bmatrix} \mathbf{u}(\mathbf{x}_i, t) \\ \dot{\mathbf{u}}(\mathbf{x}_i, t) \end{bmatrix} + \Delta t \begin{bmatrix} \dot{\mathbf{u}}(\mathbf{x}_i, t) \\ \ddot{\mathbf{u}}(\mathbf{x}_i, t) \end{bmatrix}. \tag{23}$$

Thus, $\mathcal{L}(\mathbf{U}_i^{(1)})$ can be expressed as

$$\mathcal{L}(\mathbf{U}_i^{(1)}) = \mathcal{L}\left(\begin{bmatrix} \mathbf{u}(\mathbf{x}_i, t_n) + \Delta t\dot{\mathbf{u}}(\mathbf{x}_i, t_n) \\ \dot{\mathbf{u}}(\mathbf{x}_i, t_n) + \Delta t\ddot{\mathbf{u}}(\mathbf{x}_i, t_n) \end{bmatrix}\right) = \begin{bmatrix} \dot{\mathbf{u}}(\mathbf{x}_i, t_n) + \Delta t\ddot{\mathbf{u}}(\mathbf{x}_i, t_n) \\ \mathbf{u}(\mathbf{x}_i, t_{n+1}) \end{bmatrix} \tag{24}$$

By substituting (24) into (22), we can obtain

$$\mathbf{u}(\mathbf{x}_i, t_{n+1}) = \mathbf{u}(\mathbf{x}_i, t_n) + \Delta t\dot{\mathbf{u}}(\mathbf{x}_i, t_n) + \frac{\Delta t^2}{2}\ddot{\mathbf{u}}(\mathbf{x}_i, t_n); \tag{25}$$

$$\mathbf{v}(\mathbf{x}_i, t_{n+1}) = \mathbf{v}(\mathbf{x}_i, t_n) + \frac{\Delta t}{2}(\ddot{\mathbf{u}}(\mathbf{x}_i, t_n) + \ddot{\mathbf{u}}(\mathbf{x}_i, t_{n+1})); \tag{26}$$

which is exactly the form of the VV method, a popular choice to simulate [39].

In order to obtain higher accuracy and stability, high-order methods can be used in discontinuous regions. in particular, the *third order* explicit method, *three-stage* RK method, RK3 for solving (2) can be obtain by:
(i)Stage 1:

$$\mathbf{U}_i^{(1)} = \mathbf{U}_i^n + \frac{2}{3}\Delta t \mathcal{L}_i\left(\mathbf{U}^n\right), \tag{27}$$

(ii) Stage 2:

$$\mathbf{U}_i^{(2)} = \mathbf{U}_i^n + \frac{2}{3}\Delta t \mathcal{L}_i\left(\overline{\mathbf{U}}^{(1)}\right), \tag{28}$$

(iii) Stage 3:

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n + \frac{\Delta t}{4}\left(\mathcal{L}_i(\mathbf{U}_n) + \frac{3}{2}\mathcal{L}_i\left(\mathbf{U}^{(1)}\right) + \frac{3}{2}\mathcal{L}_i\left(\mathbf{U}^{(2)}\right)\right). \tag{29}$$

Similarly, *fourth order* , *four-stage* RK method, RK4, as following:
Stage 1:

$$\mathbf{U}_i^{(1)} = \mathbf{U}_i^n + \frac{1}{2}\Delta t \mathcal{L}_i\left(\mathbf{U}^n\right), \tag{30}$$

Stage 2:

$$\mathbf{U}_i^{(2)} = \mathbf{U}_i^n + \frac{1}{2}\Delta t \mathcal{L}_i\left(\mathbf{U}^{(1)}\right), \tag{31}$$

Stage 3:

$$\mathbf{U}_i^{(3)} = \mathbf{U}_i^n + \Delta t \mathcal{L}_i\left(\mathbf{U}^{(2)}\right), \tag{32}$$

Stage 4:

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n + \frac{\Delta t}{6}\left(\mathcal{L}_i\left(\mathbf{U}^n\right) + 2\mathcal{L}_i\left(\mathbf{U}^{(1)}\right) + 2\mathcal{L}_i\left(\mathbf{U}^{(2)}\right) + \mathcal{L}_i\left(\mathbf{U}^{(3)}\right)\right). \tag{33}$$

It is noteworthy that RK3 and RK4, in comparison to RK2, permit the use of larger time steps, leading to increased efficiency in prolonged simulations. Next, we propose LTS methods based on these three methods.

4.2 Local Time-Stepping Method

For simplicity, we divide the computational domain into two subdomains Ω_C and Ω_F, where Ω_C is defined as the region with coarse time step sizes, Ω_F is defined as the region with fine time step sizes, as shown in Fig. 4:

Here $[t_n, t_{n+1}) = \cup_{k=0}^{M}[t_{n,k}, t_{n,k+1})$, where $t_{n,0} = t_n$, $t_{n,k+1} = t_{n,k} + \Delta t_f$, $\Delta t_f = \Delta t/M$ for all $k = 0, 1, 2 \ldots M$.

To initiate the LTS methods, we further partition regions Ω_C and Ω_F into four subdomains $(\Omega_{IC}, \Omega_{BC}, \Omega_{BF}, \Omega_{IF})$ as illustrated in Fig. 5. For BB-PD, subdomains Ω_{IF} and Ω_{IC} serve as internal boundary layers (BL), as depicted in Fig. 5:

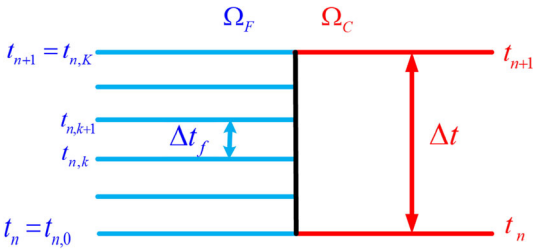


Fig. 4 Different time step sizes are utilized for the subdomains Ω_F and Ω_C, where Ω_F employs a fine time step size denoted as Δt_f, and Ω_C adopts a coarse time step size denoted as Δt

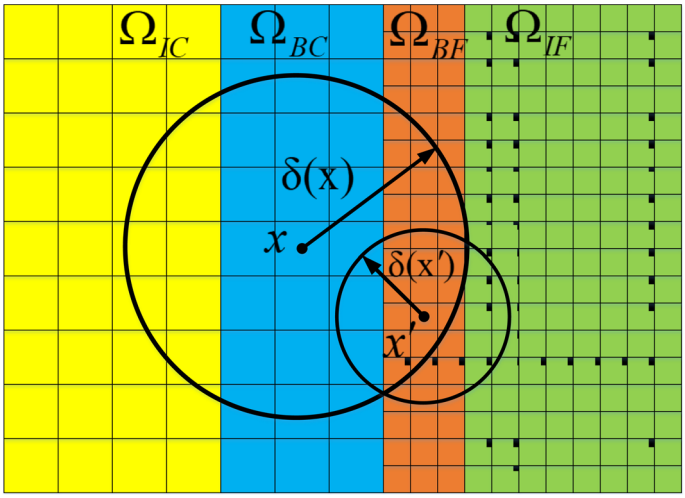


Fig. 5 The physical domain Ω is divided into four subdomains. The subdomain Ω_{IC} is depicted in yellow, while the boundary layer (BL) subdomains Ω_{BC} and Ω_{BF} are shown in blue and red, respectively. The subdomain Ω_{IF} is represented in green. Additionally, it is noted that $\Omega_C = \Omega_{BC} \cup \Omega_{IC}$ and $\Omega_F = \Omega_{BF} \cup \Omega_{IF}$

- $\Omega_{IC} = \left\{\mathbf{x}_i : \mathbf{x}_i \in \Omega_C,\ \left(\Omega_C \cap \mathcal{H}_{\mathbf{x}_i}\right) = \mathcal{H}_{\mathbf{x}_i}\right\}$.
- $\Omega_{BC} = \left\{\mathbf{x}_i : \mathbf{x}_i \in \Omega_C,\ \left(\Omega_C \cap \mathcal{H}_{\mathbf{x}_i}\right) \neq \mathcal{H}_{\mathbf{x}_i}\right\}$;
- $\Omega_{BF} = \left\{\mathbf{x}_i : \mathbf{x}_i \in \Omega_F,\ \left(\Omega_F \cap \mathcal{H}_{\mathbf{x}_i}\right) \neq \mathcal{H}_{\mathbf{x}_i}\right\}$;
- $\Omega_{IF} = \left\{\mathbf{x}_i : \mathbf{x}_i \in \Omega_F,\ \left(\Omega_F \cap \mathcal{H}_{\mathbf{x}_i}\right) = \mathcal{H}_{\mathbf{x}_i}\right\}$;

Here, Ω_{BF} and Ω_{BC} are used as boundary layers to consider the mutual influence between subregions with coarse and fine time steps. The selection of these regions is determined by the non local neighborhood size of PD.

Different numerical methods are employed based on the properties of material points within distinct subdomains in the LTS methods, specifically:

- The values $\mathbf{U}_i^n$ for all $\mathbf{x}_i \in \Omega_{IC}$, which are not influenced by the values within the subdomain Ω_{BF}, are obtained using RK methods.
- The values $\mathbf{U}_i^n$ for all $\mathbf{x}_i \in \Omega_{BC}$ are influenced by the value within the subdomain Ω_{BF}. To account for this influence, a correction term computed by the value in the previous time step is introduced.
- To compute values $\mathbf{U}_i^{n,k}$ for all $\mathbf{x}_i \in \Omega_F$, the intermediate values $\mathbf{U}_i^{n,k}$ for all $\mathbf{x}_i \in \Omega_{BC}$ are necessary, but these intermediate values are unknown. Thus, interpolation polynomials are constructed to compute these intermediate values.

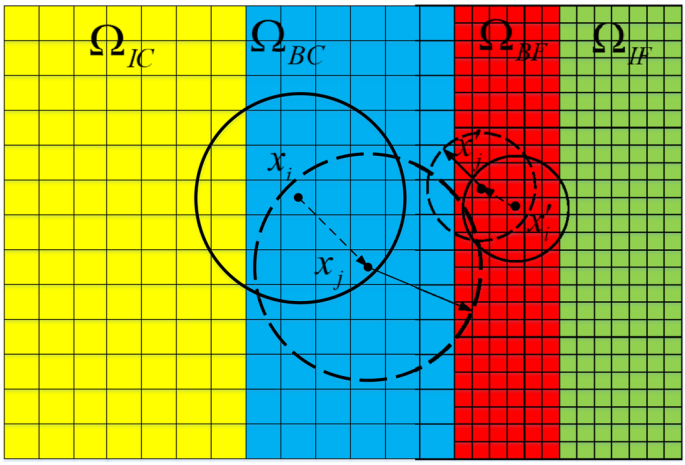


Fig. 6 The description of subdomains for OSB-PD, where the subdomain Ω_{IC} is depicted in yellow, while the boundary layer (BL) subdomains Ω_{BC} and Ω_{BF} are shown in blue and red, respectively. The subdomain Ω_{IF} is represented in green

Hereafter, we will delineate distinct algorithms for RK methods, each tailored to specific orders. Without loss of generality, let $\mathbf{U}_i^0 = (\mathbf{u}_i^0, \mathbf{v}_i^0)^{\mathrm{T}}$, where $\mathbf{u}_i^0$ and $\mathbf{v}_i^0$ is the displacement and velocity at t_0 for material point $\mathbf{x}_i$, respectively Figure6 and 10.

Remark 3 Different from BB-PD, according to (6), the peridynamics force in OSB-PD is the integral of the dilation $\theta(\mathbf{x}_j, t)$ and the displacement $\mathbf{u}(\mathbf{x}_j, t), \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}$. According to (7), the dilation $\theta(\mathbf{x}_j, t)$ is the integral of the displacement $\mathbf{u}(\mathbf{x}_k, t)$, $\mathbf{x}_k \in \mathcal{H}_{\mathbf{x}_j}$. Thus, the boundary layer should be modified as follows:

- $\Omega_{BF} = \left\{\mathbf{x}_i : \mathbf{x}_i \in \Omega_F,\ \left(\Omega_F \cap \mathcal{H}_{\mathbf{x}_j}\right) \neq \mathcal{H}_{\mathbf{x_j}},\ \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\right\}$;
- $\Omega_{BC} = \left\{\mathbf{x}_i : \mathbf{x}_i \in \Omega_C,\ \left(\Omega_C \cap \mathcal{H}_{\mathbf{x}_j}\right) \neq \mathcal{H}_{\mathbf{x_j}},\ \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\right\}$;

4.2.1 Second Order LTS Method

In this section, we introduce the second-order LTS method (LTS2) based on the RK2. To construct the LTS2, we first compute $\mathbf{U}_i^1$ using RK2. Subsequently, we proceed with the following steps:

1) Advancing the displacement from t_n to t_{n+1} on Ω_C
Let denote $\overline{\Omega}_C = \Omega_C \cup \Omega_{BF}$, $\mathbf{Q}_j^i = \mathbf{U}^i|_{\overline{\Omega}_j}$ and $\mathbf{Q}_j^{(i)} = \mathbf{U}^{(i)}|_{\overline{\Omega}_j}$. On the region Ω_{BC}, the advancement from t_n to t_{n+1} will be influenced by the material points in the region Ω_{BF}. We can estimate this impact by adding a correction term in the expression for $\mathbf{U}_i^{(1)}$, and provide the following expression for $\mathbf{U}_i^{(1)}$ on the region $\overline{\Omega}_C$:

$$\mathbf{U}_i^{(1)} = \begin{cases} \mathbf{U}_i^n + \Delta t \mathcal{L}(\mathbf{Q}_C^n), \mathbf{x}_i \in \Omega_C, & \text{(34a)} \\ \mathbf{U}_i^n + \Delta t \mathcal{L}(\mathbf{Q}_C^n) + \dfrac{\Delta t^2}{2}\left(\dfrac{\mathcal{L}_i(\mathbf{Q}_C^n) - \mathcal{L}_i(\mathbf{Q}_C^{n-1})}{\Delta t}\right), \mathbf{x}_i \in \Omega_{BF}, & \text{(34b)} \end{cases}$$

and compute $\mathbf{U}_i^{n+1}$ by RK2

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n + \frac{\Delta t}{2}\left(\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \mathcal{L}_i\left(\mathbf{Q}_C^{(1)}\right)\right),\ \mathbf{x}_i \in \Omega_C. \tag{35}$$

Then we advance the value $\mathbf{U}_i^n$ to $\mathbf{U}_i^{n+1}$ for $\mathbf{x}_i \in \Omega_C$

2) Approximate the values $\mathbf{U}_i^{n,k}$ at intermediate time levels on Ω_{BC}
At intermediate time levels $k = 1, 2, \ldots, M$, where M is the total number of intermediate time levels. The value $\mathbf{U}_i^{n,k}$ can be predicted by the values $\mathbf{U}_i^n$ and $\mathbf{U}_i^{n+1}$ through constructing an third-order interpolation polynomial as follows:

$$\begin{aligned} \mathcal{I}_i(t) &= \mathbf{U}_i^n + (t - t_n)\,\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + (t - t_n)^2\,\beta,\ \ t_n \le t \le t_{n+1}, \\ \beta &= \frac{\mathbf{U}_i^{n+1} - \mathbf{U}_i^n - \Delta t\,\mathcal{L}_i\left(\mathbf{Q}_C^n\right)}{\Delta t^2},\ \ \mathbf{x}_i \in \Omega_{BC}, \end{aligned} \tag{36}$$

Then the value $\mathbf{U}_i^{n,k}$, $\mathbf{x}_i \in \Omega_{BC}$ can be approximated by the value $\mathcal{I}_i(t_{n,k})$.

3) Advancing the displacement from t_n to t_{n+1} on Ω_F
Let $\overline{\Omega}_F = \Omega_{BC} \cup \Omega_F$, then we can compute the solution as follows:

i) Stage 1:

$$\mathbf{U}_i^{(1,k)} = \begin{cases} \mathbf{U}_i^{n,k} + \Delta t_f\,\mathcal{L}_i\left(\mathbf{Q}_F^{n,k}\right),\ \ \mathbf{x}_i \in \Omega_F, & \text{(37a)} \\ \mathbf{U}_i^{n,k} + \Delta t_f\,\mathcal{I}_i^{'}(t_{n,k}),\ \ \mathbf{x}_i \in \Omega_{BC}, & \text{(37b)} \end{cases}$$

ii) Stage 2:

$$\mathbf{U}_i^{n,k+1} = \mathbf{U}_i^{n,k} + \frac{\Delta t_f}{2}\left(\mathcal{L}_i\left(\mathbf{Q}_F^{n,k}\right) + \mathcal{L}_i\left(\mathbf{Q}_F^{(1,k)}\right)\right), \mathbf{x}_i \in \Omega_F. \tag{38}$$

Then we obtain $\mathbf{U}_i^{n+1}$ for $\mathbf{x}_i \in \Omega = \Omega_C \cup \Omega_F$. The proposed LTS2 satisfies the following theorem:

Theorem 1 *if $\mathcal{L}_i$ for $\mathbf{x}_i \in \Omega$ is C^3, the global error of the LTS2 is $O(\Delta t^2)$.*

The proof can be found in the Appendix A.

4.2.2 Third Order LTS Method

This section introduces a third-order LTS method (LTS3) based on third-order RK to address the problem with different time step sizes. Obtain the value $\mathbf{U}_i^1$ using RK3, and then we have the following steps:

1) Advancing displacement from t_n to t_{n+1} on Ω_C
On the the subdomain Ω_C, we can compute the value $\mathbf{U}_i^{(1)}$ by follows

$$\mathbf{U}_i^{(1)} = \mathbf{U}_i^n + \frac{2}{3}\Delta t\,\mathcal{L}_i\left(\mathbf{Q}_C^n\right),\ \ \mathbf{x}_i \in \overline{\Omega}_C. \tag{39}$$

On the subdomain Ω_{BC}, the advancement from t_n to t_{n+1} is influenced by the material points in subdomain Ω_{FI}. This influence can be estimated by incorporating a correction term in the expression for $\mathbf{U}_i^{(2)}$. We can then provide the following expression for $\mathbf{U}_i^{(2)}$

on the subdomain $\overline{\Omega}_C$:

$$\mathbf{U}_i^{(2)} = \begin{cases} \mathbf{U}_i^n + \dfrac{2}{3}\Delta t \mathcal{L}_i\left(\mathbf{Q}_C^{(1)}\right), \ \mathbf{x}_i \in \Omega_C, & \text{(40a)} \\ \mathbf{U}_i^n + \dfrac{2}{3}\Delta t \mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \dfrac{4}{9}\Delta t^2 \left(\dfrac{\mathcal{L}_i\left(\mathbf{Q}_C^n\right) - \mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}\right), \mathbf{x}_i \in \Omega_{BF}, & \text{(40b)} \end{cases}$$

then the value $\mathbf{U}_i^{n+1}, \mathbf{x}_i \in \Omega_C \cap \Omega_{BC}$ can be obtained by follows:

$$\mathbf{U}_i^{(1)} = \mathbf{U}_i^n + \frac{\Delta t}{4}\left(\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \mathcal{L}_i\left(\mathbf{Q}_C^{(1)}\right) + \mathcal{L}_i\left(\mathbf{Q}_C^{(2)}\right)\right), \ \mathbf{x}_i \in \Omega_C. \tag{41}$$

Therefore, we advance from t_n to t_{n+1} on the subdomain Ω_C.

2) Approximate the values $\mathbf{U}_i^{n,k}$ at intermediate time levels on Ω_{BC}
At intermediate time levels $k = 1, 2, \ldots, M$, where M is the total number of intermediate time levels . The value $\mathbf{U}_i^{n,k}$ can be predicted by the values $\mathbf{U}_i^n$ and $\mathbf{U}_i^{n+1}$ through constructing an third-order interpolation polynomial as follows:

$$\begin{aligned} \mathcal{I}_i(t) = \mathbf{U}_i^n + (t - t_n)\,\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + (t - t_n)^2(\beta - \Delta t\gamma) \\ + \frac{(t - t_n)^3}{6}\beta, \ t_n \le t \le t_{n+1}, \ \mathbf{x}_i \in \Omega_{BC}. \end{aligned} \tag{42}$$

Here, γ is a parameter to ensure that the local error of the interpolation function is fourth-order. Using finite differences, we have

$$\gamma = \frac{6}{5\Delta t}\left(2\frac{\mathbf{U}_i^{n+1} - \mathbf{U}_i^n - \Delta t\mathcal{L}_i\left(\mathbf{Q}_C^n\right)}{\Delta t^2} - \frac{\mathcal{L}_i\left(\mathbf{Q}_C^n\right) - \mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}\right). \tag{43}$$

Then the value $\mathbf{U}_i^{n,k}, \mathbf{x}_i \in \Omega_{BC}$ can be approximated by the value $\mathcal{I}_i(t_{n,k})$.

3) Advancing displacement from t_n to t_{n+1} on Ω_F
Let $\overline{\Omega}_F = \Omega_{BC} \cup \Omega_F$, then we can compute the solution using the RK3 with the fine time step size as follows:

i) Stage 1:

$$\mathbf{U}_i^{(1,k)} = \begin{cases} \mathbf{U}_i^{n,k} + \dfrac{2\Delta t_f}{3}\mathcal{L}_i\left(\mathbf{Q}_F^{n,k}\right), \ \mathbf{x}_i \in \Omega_F, & \text{(44a)} \\ \mathbf{U}_i^{n,k} + \dfrac{2\Delta t_f}{3}\mathcal{I}_i'(t_{n,k}), \ \mathbf{x}_i \in \Omega_{BC}, & \text{(44b)} \end{cases}$$

ii) Stage 2:

$$\mathbf{U}_i^{(2,k)} = \begin{cases} \mathbf{U}_i^{n,k} + \dfrac{2\Delta t_f}{3}\mathcal{L}_i\left(\mathbf{Q}_F^{(1,k)}\right), \ \mathbf{x}_i \in \Omega_F, & \text{(45a)} \\ \mathbf{U}_i^{n,k} + \dfrac{2\Delta t_f}{3}\mathcal{I}_i'(t_{n,k}) + \dfrac{4\Delta t_f}{9}\mathcal{I}_i''(t_{n,k}), \ \mathbf{x}_i \in \Omega_{BC}, & \text{(45b)} \end{cases}$$

iii) Stage 3:

$$\begin{aligned} \mathbf{U}_i^{n,k+1} = &\mathbf{U}_i^{n,k} + \frac{\Delta t_f}{4}\left(\mathcal{L}_i\left(\mathbf{Q}_F^{n,k}\right) + \frac{3}{2}\mathcal{L}_i\left(\mathbf{Q}_F^{(1,k)}\right) + \frac{3}{2}\ \mathcal{L}_i\left(\mathbf{Q}_F^{(2,k)}\right)\right), \\ &\mathbf{x}_i \in \Omega_F. \end{aligned} \tag{46}$$

The proposal satisfy the following theorem:

Theorem 2 *if $\mathcal{L}_i$ for $\mathbf{x}_i \in \Omega$ is C^4, the global error of the LTS3 is $O(\Delta t^3)$.*

The proof can be found in Appendix B.

4.2.3 Fourth-Order LTS Method

The fourth-order LTS method (LTS4) based on RK4 is introduced similarly to the third-order method. Compute the value $\mathbf{U}(\mathbf{x}_i, t_1)$ and $\mathbf{U}(\mathbf{x}_i, t_2)$ by RK4, and then we perform the following steps to implement the LTS4:

1) Advancing displacement from t_n to t_{n+1} on Ω_C
First we compute the value $\mathbf{U}_i^{(1)}$ as follows:

$$\mathbf{U}_i^{(1)} = \mathbf{U}_i^n + \frac{1}{2}\Delta t \mathcal{L}_i\left(\mathbf{Q}_C^n\right), \ \mathbf{x}_i \in \overline{\Omega}_C, \tag{47}$$

Then we can obtain $\mathbf{U}_i^{(2)}$ and $\mathbf{U}_i^{(3)}$ by the following equation:

$$\mathbf{U}_i^{(2)} = \begin{cases} \mathbf{U}_i^n + \frac{1}{2}\Delta t \mathcal{L}_i\left(\mathbf{Q}_n^{(1)}\right), \ \mathbf{x}_i \in \Omega_C, & \text{(48a)} \\ \mathbf{U}_i^n + \frac{1}{2}\Delta t \mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \frac{1}{4}\Delta t^2 \left(\frac{\mathcal{L}_i\left(\mathbf{Q}_C^n\right) - \mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}\right), \ \mathbf{x}_i \in \Omega_{BF}, & \text{(48b)} \end{cases}$$

and

$$\mathbf{U}_i^{(3)} = \begin{cases} \mathbf{U}_i^n + \Delta t \mathcal{L}_i\left(\mathbf{Q}_n^{(2)}\right), \ \mathbf{x}_i \in \Omega_C, & \text{(49a)} \\ \mathbf{U}_i^n + \Delta t \mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \frac{1}{2}\Delta t^2 \left(\frac{\mathcal{L}_i\left(\mathbf{Q}_C^n\right) - \mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}\right) + \\ \frac{3\Delta t^2}{4}\left(\frac{\mathcal{L}_i\left(\mathbf{Q}_C^n\right) - \mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t} - \frac{\mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right) - \mathcal{L}_i\left(\mathbf{Q}_C^{n-2}\right)}{\Delta t}\right), \ \mathbf{x}_i \in \Omega_{BF}, & \text{(49b)} \end{cases}$$

and $\mathbf{U}_i^{n+1}, \mathbf{x}_i \in \Omega_F$ can be obtained by follows

$$\mathbf{U}_i^{(1)} = \mathbf{U}_i^n + \frac{\Delta t}{6}\left(\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + 2\mathcal{L}_i\left(\mathbf{Q}_C^{(1)}\right) + 2\mathcal{L}_i\left(\mathbf{Q}_C^{(2)}\right) + 2\mathcal{L}_i\left(\mathbf{Q}_C^{(3)}\right)\right), \ \mathbf{x}_i \in \Omega_C. \tag{50}$$

2) Approximate the values $\mathbf{U}_i^{n,k}$ at intermediate time levels on Ω_{BC}
We can conduct an interpolation polynomial as follows:

$$\begin{aligned} \mathcal{I}_i(t) =& \mathbf{U}_i^n + (t - t_n)\,\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + (t - t_n)^2\left(\beta - \frac{\Delta t}{6}\epsilon + \frac{\Delta t^2}{8}\theta\right) \\ &+ \frac{(t - t_n)^3}{6}(\beta - \Delta t\theta) + \frac{(t - t_n)^4}{24}\theta, \quad t_n \le t \le t_{n+1}, \end{aligned} \tag{51}$$

here γ and ϵ are parameters to ensure that the local error of the interpolation function is fifth-order, which is defined by

$$\epsilon = \frac{6}{5\Delta t}\left(2\frac{\mathbf{U}_i^{n+1}-\mathbf{U}_i^n-\Delta t\mathcal{L}_i\left(\mathbf{Q}_C^n\right)}{\Delta t^2}-\frac{\mathcal{L}_i\left(\mathbf{Q}_C^n\right)-\mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}\right)+\frac{11\Delta t}{10}\theta, \tag{52}$$

and

$$\begin{aligned}\theta =&\frac{10}{7\Delta t}\Bigg[\frac{6}{5\Delta t}\left(2\frac{\mathbf{U}_i^{n+1}-\mathbf{U}_i^n-\Delta t\mathcal{L}_i\left(\mathbf{Q}_C^n\right)}{\Delta t^2}-\frac{\mathcal{L}_i\left(\mathbf{Q}_C^n\right)-\mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}\right)\\&-\frac{1}{\Delta t}\left(\frac{\mathcal{L}_i\left(\mathbf{Q}_C^n\right)-\mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}-\frac{\mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)-\mathcal{L}_i\left(\mathbf{Q}_C^{n-2}\right)}{\Delta t}\right)\Bigg],\end{aligned}$$

then $\mathbf{U}_i^{n,k}$ can be approximated by $\mathcal{I}_i(t_{n,k})$, $k=1,2\ldots K$ for $\mathbf{x}_i\in\Omega_{BC}$.

3) Advancing displacement from t_n to t_{n+1} on Ω_F At each intermediate time level $k=1,2,\ldots,M$, the values with the fine time step size can be computed as follows:

i) Stage 1:

$$\mathbf{U}_i^{(1,k)}=\begin{cases}\mathbf{U}_i^{n,k}+\dfrac{\Delta t_f}{2}\mathcal{L}_i\left(\mathbf{Q}_F^{n,k}\right), & \mathbf{x}_i\in\Omega_F, \quad (53a)\\ \mathbf{U}_i^{n,k}+\dfrac{\Delta t_f}{2}\mathcal{I}_i^{'}(t_{n,k}), & \mathbf{x}_i\in\Omega_{BC}, \quad (53b)\end{cases}$$

ii) Stage 2:

$$\mathbf{U}_i^{(2,k)}=\begin{cases}\mathbf{U}_i^{n,k}+\dfrac{\Delta t_f}{2}\mathcal{L}_i\left(\mathbf{Q}_F^{(1,k)}\right), & \mathbf{x}_i\in\Omega_F, \quad (54a)\\ \mathbf{U}_i^{n,k}+\dfrac{\Delta t_f}{2}\mathcal{I}_i^{'}(t_{n,k})+\dfrac{\Delta t_f^2}{4}\mathcal{I}_i^{''}(t_{n,k}), & \mathbf{x}_i\in\Omega_{BC}, \quad (54b)\end{cases}$$

iii) Stage 3:

$$\mathbf{U}_i^{(3,k)}=\begin{cases}\mathbf{U}_i^{n,k}+\dfrac{\Delta t_f}{2}\mathcal{L}_i\left(\mathbf{Q}_F^{(2,k)}\right), & \mathbf{x}_i\in\Omega_F, \quad (55a)\\ \mathbf{U}_i^{n,k}+\dfrac{\Delta t_f}{2}\mathcal{I}_i^{'}(t_{n,k})+\dfrac{\Delta t_f^2}{2}\mathcal{I}_i^{''}(t_{n,k})+\dfrac{\Delta t_f^3}{4}\mathcal{I}_i^{'''}(t_{n,k}), & \mathbf{x}_i\in\Omega_{BC}, \quad (55b)\end{cases}$$

iv) Stage 4:

$$\begin{aligned}\mathbf{U}_i^{n+1}=&\mathbf{U}_i^n+\frac{\Delta t}{6}\left(\mathcal{L}_i\left(\mathbf{Q}_F^{n,k}\right)+2\mathcal{L}_i\left(\mathbf{Q}_F^{(1,k)}\right)+2\mathcal{L}_i\left(\mathbf{Q}_F^{(2,k)}\right)+\mathcal{L}_i\left(\mathbf{Q}_F^{(3,k)}\right)\right),\\&\mathbf{x}_i\in\Omega_F.\end{aligned} \tag{56}$$

The proposed LTS4 satisfies the following theorem:

Theorem 3 *if $\mathcal{L}_i$ for $\mathbf{x}_i\in\Omega$ is C^5, the global error of the LTS4 is $O(\Delta t^4)$.*

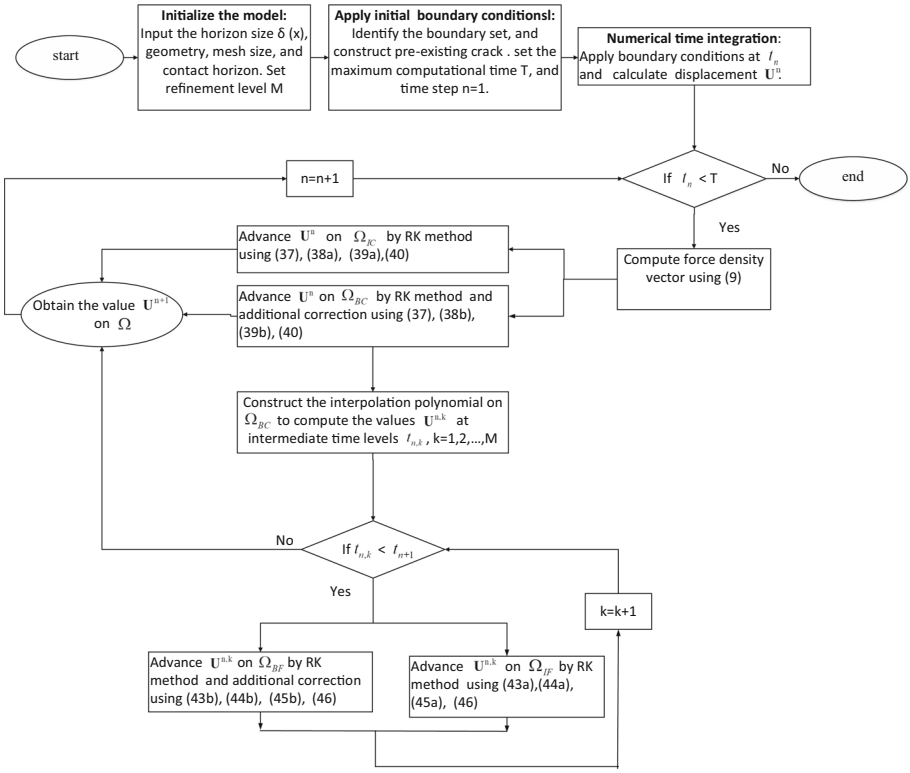


Fig. 7 The flowchart of the LTS methods

The proof can be found in Appendix C. For a visual representation of the proposed algorithm, refer to the flowchart in Fig. 7. Here, we take the LTS4 as an example, and the process for the third and second orders is the same.

For state-based peridynamics, we can use the MTS method mentioned in Section 4 for temporal discretization. However, it should be noted that in the state-based model, the boundary region Ω_{BC} and Ω_{BF} needs to be selected as 2δ instead of δ to ensure that all interacting material points are selected.

5 Numerical Example

In this section, we showcase various numerical examples to illustrate the accuracy and convergence of the LTS method for BB-PD across different geometries. All experiments were performed using Matlab® on a workstation equipped with an Intel Xeon Gold 6240 CPU operating at 2.6GHz, and 2048GB of installed memory.

5.1 Courant-Friedrichs-Lewy Condition Test

For numerical experiments, we begin by analyzing a plate subjected to uniaxial tension. The goal is to verify that the CFL condition of the PD is primarily determined by the horizon size

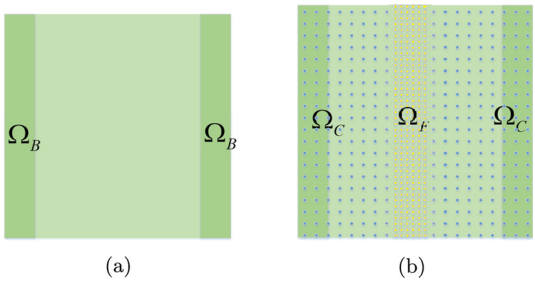


Fig. 8 The geometric representation of a 2D plate alongside its discretization

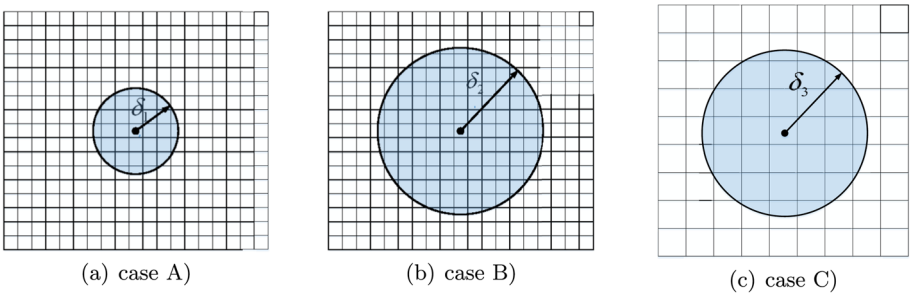


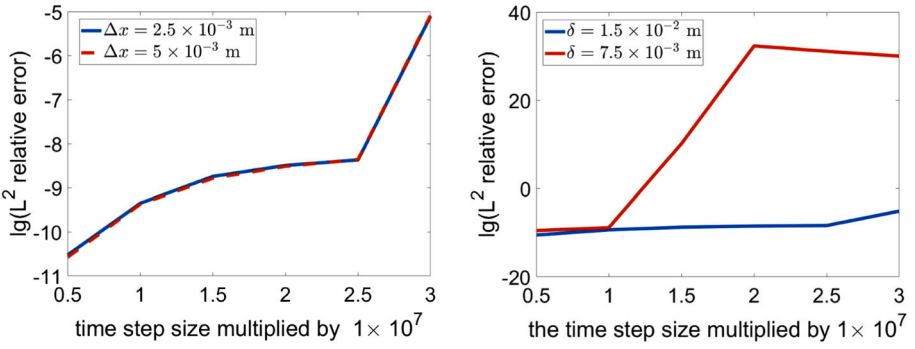
Fig. 9 Comparison of δ and Δx under three spatial discretizations, where $\delta_3 = \delta_2 = 2\delta_1 = 3\Delta x_3 = 6\Delta x_2 = 6\Delta x_1 = 7.5 \times 10^{-3}$ m

rather than the grid spacing. The plate has dimensions of $L = 0.5$ m (length), $W = 0.5$ m (width), and $h = 0.01$ m (thickness). Fig. 8 (a) depicts the geometry of the plate. The material properties are $\rho = 8000\ \text{kg/m}^3$ (density), $\nu = 1/3$ (Poisson's ratio) and $E = 1.92 \times 10^{11}$ Mpa (elastic modulus). $b_p = p_0 W/h$ represents the external loading which is applied to the boundary layer Ω_C, where $p_0 = 200$ Mpa is the uniaxial tension loading, and the width of Ω_B is 0.01 m.

To explore the influence of CFL conditions, we consider three different cases: In case A), the horizon is chosen as $\delta_1 = 3\Delta x_1 = 7.5 \times 10^{-3}$ m, as shown in Fig. 9 (a); In case B), the horizon is chosen as $\delta_2 = 6\Delta x_2 = 1.5 \times 10^{-2}$ m, as shown in Fig. 9 (b); In case C), the horizon is chosen as $\delta_3 = 3\Delta x_3 = 1.5 \times 10^{-2}$ m, as shown in Fig. 9 (c).

We investigate the influence of horizon size on CFL conditions through cases A) and B), maintaining identical grid spacing but differing in horizon sizes. Case C) facilitates comparison under consistent horizon size but varying grid spacing, confirming that CFL conditions remain unaffected by changes in grid spacing. Simulations utilize RK4 with a uniform total time of $6 \times 10^{-6} s$ for each case. L^2 relative error is computed using a reference solution obtained with RK4 at a very small time step of $\Delta t = t = 1 \times 10^{-9} s$. Plots illustrating numerical errors for various time steps are presented in Fig. 11.

We observe that among the three cases considered, Fig.11 (b) exhibits the poorest performance. When the time step size exceeds $1 \times 10^{-7} s$, there is a notable increase in computational error. As anticipated, the calculation errors in Fig.11 (a) are very similar. This emphasizes that the horizon is the decisive factor influencing changes in CFL conditions.



(a) (b)

Fig. 10 L^2 relative error obtained at various time step sizes: (a) Error variation with same δ and different Δx; (b) Error variation with same Δx and different δ

5.2 Convergence Rate Test

In this section, we investigate the convergence of the proposed LTS method to verify the accuracy. The model used here is consistent with section 5.1.

To assess the accuracy and computational efficiency of algorithms under local refinement, we divide the subdomain into two distinct regions: $\Omega_F = [-L/5, L/5] \times [-W/2, W/2]$ and $\Omega_C = [-L/2, -L/5] \times [-W/2, W/2] \cup [L/5, L/2] \times [-W/2, W/2]$, as illustrated in Figure 8 (b). In Ω_C, we apply a coarse time step size Δt_c and a coarse grid spacing Δx_c. Conversely, in Ω_F, we employ a fine time grid with Δt_f and a fine spatial step size Δx_f. Based on this division, we disregard spatial errors to evaluate the temporal convergence order for the LTS method, selecting

$$\frac{\Delta t_c}{\Delta t_f} = M, \quad \frac{\Delta x_c}{\Delta x_f} = M, \quad \Delta x_f = 1.25 \times 10^{-3}\text{m}. \tag{57}$$

Following this, we conducted an assessment to gauge the convergence across various values of M for the LTS methods. Throughout this analysis, we changed the coarse time step size and calculated the convergence at different ratios $M = \{2, 4, 8\}$. We calculated the L^2 relative error at $T = 4 \times 10^{-6}$ s, utilizing the classical RK2, RK3, RK4 reference solution obtained with a highly refined time step size of $\Delta t = t = 1 \times 10^{-9}$ s for LTS2, LTS3, LTS4, respectively. Comprehensive results are outlined in Table. 1. From this Table, we conclude that LTS methods reach their respective convergence order.

Compared to the LTS2, the LTS3 and LTS4 permit larger time step sizes, resulting in comparable accuracy achieved within shorter durations. Table 2 demonstrates that for equal errors, LTS4 and LTS3 require shorter CPU times, highlighting the increased efficiency of the proposed higher-order algorithm over lower-order alternatives.

5.3 Boundary Handling Test

To further verify the effectiveness of our algorithm, we consider a multi-time-stepping method without interface treatment, in which the fine time-step values $U_{n,k}$ are computed directly

Table 1 L^2 relative error between the RK reference solution with fine time step size $\Delta t = t = 1 \times 10^{-9}$ and the approximate solutions obtained by LTS2, LTS3, LTS4. CR. represents the rates of convergence

		LTS2		LTS3		LTS4	
Δt_c	M	error	CR	error	CR	error	CR
3.125E−8	2	1.58E-5		4.66E-8	–	7.66E-10	–
	4	1.58E-5		4.66E-8	–	7.64E-10	–
	8	1.57E-5		4.65E-8	–	7.64E-10	–
6.25E−8	2	8.15E-5	1.99	3.71E-7	3.02	1.96E-9	3.83
	4	8.14E-5	1.99	3.71E-7	3.02	1.96E-9	3.83
	8	8.14E-5	1.99	3.70E-7	3.01	2.95E-9	3.83
1.25E−7	2	2.06E-4	1.97	2.92E-6	3.01	2.86E-8	3.93
	4	2.06E-4	1.98	2.92E-6	3.01	2.86E-8	3.93
	8	2.05E-5	1.97	2.91E-6	3.01	2.85E-8	3.93

Table 2 CPU times obtained by LTS2, LTS3, LTS4 under same L^2 relative error. Here the ratio $M = 2$

	LTS2	LTS3	LTS4
Δt_c	$8 \times 10^{-8} s$	$4 \times 10^{-7} s$	$5.4 \times 10^{-6} s$
error	2.47E-5	2.45E-5	2.43E-5
CPU times	55m28s	31m29s	21m20s

from U_{n+1} rather than obtained from an interpolation function. The example used is the same as that in Section 5.1.

To evaluate the accuracy and efficiency of the algorithms under local refinement, the subdomain is partitioned into two separate regions: $\Omega_F = [-L/5, L/5] \times [-W/2, W/2]$ and $\Omega_C = [-L/2, -L/5] \times [-W/2, W/2] \cup [L/5, L/2] \times [-W/2, W/2]$, as depicted in Figure 8 (b). In Ω_C, a coarse time step size Δt_c and a coarse spatial resolution Δx_c are utilized, whereas in Ω_F, a fine time step Δt_f and fine spatial discretization $\Delta x_f = 2.5 \times 10^{-3}$m are adopted.

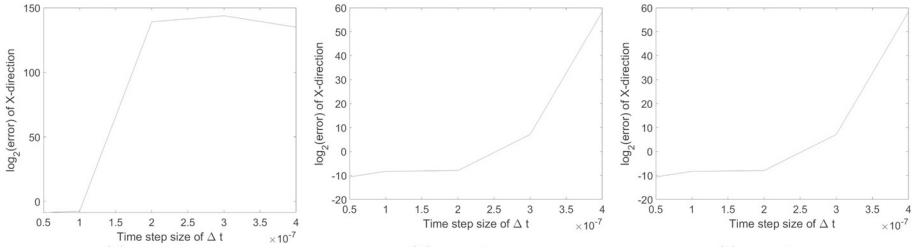
Following this, we conducted an assessment to examine the convergence behavior of the LTS methods across different values of M. In this analysis, we varied the coarse time step size and evaluated convergence at $M = 4$. The L^2 relative error at $T = 4 \times 10^{-7}$ s was computed using classical RK2, RK3, and RK4 reference solutions, obtained with a highly refined time step of $\Delta t = t = 1 \times 10^{-10}$ s for LTS2, LTS3, and LTS4, respectively. A detailed summary of the results is presented in Table 3.It can be observed that the traditional approach without interpolation functions is highly unstable, achieving satisfactory accuracy only with third-order schemes, and none of the methods attain the expected order of convergence. From this table, it is evident that the correct order of convergence cannot be achieved without boundary treatment, highlighting its essential role in ensuring the accuracy of the method.

5.4 Stability for Different Time-Step Test

In this section, we investigate the effect of time-step selection on stability in different regions. We consider three cases:

Table 3 L^2 relative error between the RK reference solution with fine time step size $\Delta t = t = 1 \times 10^{-9}$ and the approximate solutions obtained by LTS2, LTS3, LTS4. CR. represents the rates of convergence, Δt_c represent the coarse time step

Δt_c	M	LTS2 error	CR	LTS3 error	CR	LTS4 error	CR
1E−8	2	4.24E-04	-	1.15E-12	-	7.48E-04	-
5E−9	2	1.84E-04	1.20	1.98E-13	2.53	5.29E-04	0.49
2.5E−9	2	9.13E-05	1.04	5.64E-14	1.82	3.06E-04	0.78



(a) case 1 (b) case 2 (c) case 3

Fig. 11 L^2 relative error obtained at various cases

case 1: A uniform coarse grid is used for both space and time, with spatial resolution $\Delta x = 2.5 \times 10^{-3} m$, and the corresponding coarse time step is applied throughout the domain.

case 2: A uniform coarse grid is used for both space and time, with spatial resolution $\Delta x = 1.25 \times 10^{-3} m$, and the corresponding coarse time step is applied throughout the domain.

case 3: A coarse-fine coupled grid is employed, as described in Section 5.2, where the fine region uses $\Delta x_f = 1.25 \times 10^{-3} m$ and the coarse region uses $\Delta x = 1.25 \times 10^{-3} m$. The spatial-to-temporal resolution ratio is set to $M = 2$, meaning that the fine time step Δt_f is half the coarse one Δt.

In all cases, we set $\delta = 3\Delta x$ and $\delta_f = 3\Delta x_f$. The time-step size is increased from 2.5×10^{-4} to 1×10^{-3}, while all other parameters remain the same as those in Section 5.3. The resulting L^2 errors obtained by the LTS4 method are summarized as follows:

It can be observed that for larger time-step sizes, the fine spatial grid becomes unstable, whereas both the coarse spatial grid and the proposed multi-resolution configuration remain stable. This confirms the stability of the proposed algorithm when employing different time-step sizes in different regions.

5.5 Simulation of Plate with Pre-Existing Crack using Bond-Based Peridynamics

We further examine the performance of our LTS methods using a plate with a pre-existing crack to verify the accuracy on 2D non continuous problem. Illustrated in Fig. 12 (a) is the 2D plate under consideration, with dimensions $L = 0.05$ m (length), $W = 0.05$ m (width), $h = 0.01$ m (thickness). A pre-existing crack of length $2c = 0.01$ m is present within the plate. Material properties such as elastic modulus E, horizon size δ, density ρ, and Poisson's ratio μ align with those specified in Section 5.1. Furthermore, velocity constraints are imposed

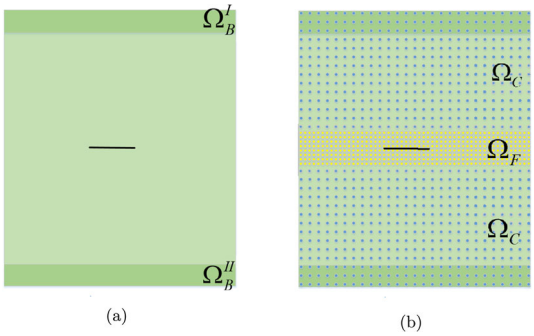


Fig. 12 The geometric configuration of a 2D plate featuring a pre-existing crack

Table 4 CPU times for solving the plate with pre-existing crack, using LTS2, LTS3, and LTS4 with DH-PD

N	M	LTS2	LTS3	LTS4
2000	1	44m41s	59m48s	1h18m
1000	2	21m31s	28m21s	35m10s
500	4	15m52s	22m28s	27m37s
250	8	10m26s	16m20s	21m10s

on Ω_B^1 and Ω_B^{II} at the top and bottom of subdomain Ω, respectively, each with a depth of δ.

$$\begin{aligned} \dot{u}_y(\mathbf{x}_i, t) &= \quad 20.0\ \text{m/s} \quad \mathbf{x}_i \in \Omega_B^I, \\ \dot{u}_y(\mathbf{x}_i, t) &= -20.0\ \text{m/s} \quad \mathbf{x}_i \in \Omega_B^{II}. \end{aligned} \tag{58}$$

In order to implement the LTS method, we decompose the region $\Omega = [-L/2, L/2] \times [-W/2, W/2]$ into two distinct subdomains: $\Omega_F = [-L/2, L/2] \times [-W/5, W/5]$ and $\Omega_C = [-L/2, L/2] \times [-W/2, -W/5] \cup [-L/2, L/2] \times [W/5, W/2]$, as shown in Fig. 12 (b). In Ω_F, characterized as a discontinuous region, we employ fine time steps with $\Delta t_f = 1.667 \times 10^{-3}$s and spatial grids with $\Delta x_f = 1 \times 10^{-4}$m. On the other hand, Ω_C is identified as a smooth region, governed by coarse time steps with $\Delta t_c = M \Delta t_f$ and spatial grids with $\Delta x_c = M \Delta x_f$. According to [40], the fracture constant s_0 is influenced by the horizon size δ, which means This means that for different subdomains with different horizons, different values of s_0 should be adopt. Thus, to determine the fracture, we introduce s_0 as follows

$$s_0(\mathbf{x}, \mathbf{x}', t) = \begin{cases} 4.472 \times 10^{-2}, & \mathbf{x} \in \Omega_C, \\ M \times 4.472 \times 10^{-2}, & \mathbf{x} \in \Omega_F \end{cases} \tag{59}$$

Following this, our objective is to assess the inherent computational efficiency advantages of the LTS method compared to the RK method. Keeping the physical time $T = 1.667 \times 10^{-1} s$, fine time step size $\Delta t_f = 1.667 \times 10^{-3}$s, and fine grid spacing $\Delta x_f = 1 \times 10^{-4}$m fixed, we using various values of $M \in \{1, 2, 4, 8\}$ and compared and analyzed the CPU time. It's important to note that when $M = 1$, the proposed LTS method and the RK method are consistent. Detailed data can be found in Table 4, where N represents the subintervals in the subdomain Ω_C. It's worth noting that the RK method necessitates more time steps to achieve results consistent with the LTS method.

Next, we verify the accuracy of the LTS method in addressing crack propagation. Here we choose two methods for illustration: (a) the global time-stepping method, ie., RK4; (b) LTS4 with $M = 8$. The simulation outputs are depicted in Fig. 13. In this visualization,

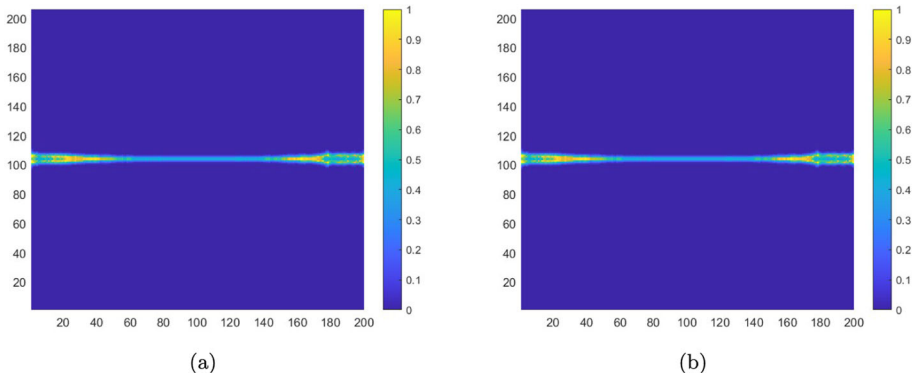


Fig. 13 Crack propagation simulation with different methods: (a) RK4; (b) LTS4 with $M = 8$

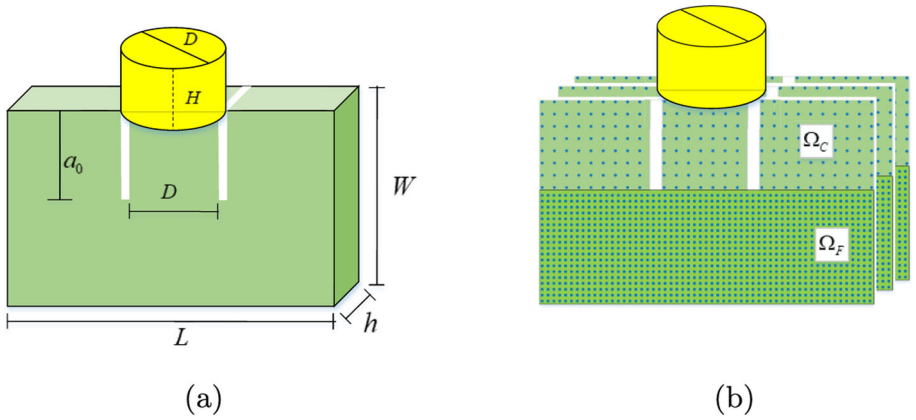


Fig. 14 The geometry of the Kalthoff-Winkler experiment, along with its discretization

blue regions signify intact bonds with $s = 1$, while yellow regions denote bond fracture with $s = 0$. Regions marked by substantial material point stretching exhibit values of s spanning from 0 to 1, reflecting instances of partial bond failure. These results confirm the LTS method's proficiency in capturing crack propagation within the same physical time, akin to the outcomes derived from the RK method.

5.6 Simulation of 3D Model with Pre-Existing Crack using the Bond-Based Peridynamics

To showcase the effectiveness of the LTS method in a 3D fracture model, we conducted the Kalthoff-Winkler experiment, as depicted in Fig. 14.

The problem entails a block measuring $L = 0.2$ m in length, $W = 0.1$ m in width, and $h = 0.009$ m in thickness, featuring two thin notches. In this model, each notch has a width of $h_0 = 0.0015$ m, a length of $a_0 = 0.05$ m, and a distance between notches of $d = 0.05$ m. The impactor possesses a diameter of $D = 0.05$ m and a height of $H = 0.05$ m. The material properties are elastic modulus $E = 1.91 \times 10^9 Mpa$, Poisson's ratio $\upsilon = 1/4$, horizon size $\delta = 3 \times 10^{-4}$ and density $\rho = 8000\ \mathrm{kg/m^3}$. An initial velocity of $v_0 = 32$ m/s is enforced

Table 5 CPU times for solving the Kalthoff-Winkler experiment, using LTS2, LTS3 and LTS4 with DH-PD

N	M	LTS2	LTS3	LTS4
1400	1	12h30m	19h10m	23h48m
700	2	7h20m	11h51m	14h32m
350	4	6h12m	9h42m	12h48m
175	8	5h23m	8h18m	10h54m

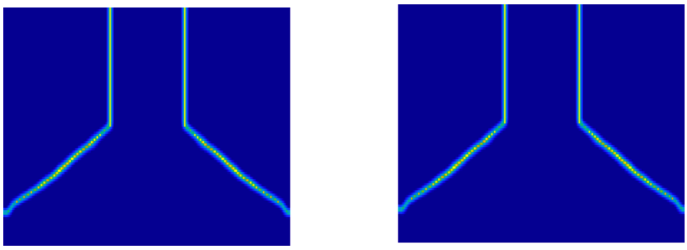


Fig. 15 Crack propagation simulation using LTS4 with different methods: (a) RK4; (b) LTS4 with $M = 8$

to facilitate crack propagation. The critical stretch is defined as

$$s_0(\mathbf{x}, \mathbf{x}', t) = \begin{cases} 1 \times 10^{-2}, & \mathbf{x} \in \Omega_C, \\ M \times 10^{-2}, & \mathbf{x} \in \Omega_F \end{cases} \tag{60}$$

We decompose the region $\Omega = [-L/2, L/2] \times [-W/2, W/2] \times [-h/2, h/2]$ into two distinct subdomains: $\Omega_C = [-L/2, L/2] \times [-W/5, W/5] \times [-h/2, h/2]$ and $\Omega_F = [-L/2, L/2] \times [-W/2, -W/5] \times [-h/2, h/2] \cup [-L/2, L/2] \times [W/5, W/2] \times [-h/2, h/2]$, as shown in Fig. 12 (b). In Ω_F, characterized as a discontinuous region, we employ fine time steps with $\Delta t_f = 8.7 \times 10^{-8}$ s and spatial grids with $\Delta x_f = 1 \times 10^{-4}$ m. On the other hand, Ω_C is identified as a smooth region, governed by coarse time steps with $\Delta t_c = M \Delta t_f$ and spatial grids with $\Delta x_c = M \Delta x_f$, where M is a constant ensuring synchronization between the time and spatial discretizations in the coarse region.

The CPU time statistics are presented in Table 5, where we maintain the physical time at $T = 1.21 \times 10^{-4} s$ and assess the performance of three methods under varying parameters M when Δt_f and Δx_f remains unchanged. The results demonstrate that our proposed method maintains high computational efficiency in three-dimensional fracture problems. Notably, the improvement in computational efficiency is contingent on the ratio of discontinuous to smooth regions. In this specific scenario, the chosen proportion of discontinuous regions is larger compared to two-dimensional examples, resulting in relatively lower computational time savings. For three-dimensional problems with more minor crack propagation, the enhancement in computational efficiency would be more pronounced.

To verify the accuracy, we choose two examples using different methods and same cross-sections to depict crack propagation, as depicted in Fig. 15. Both examples exhibit a consistent crack angle of 70 degrees, mirroring results observed in physical experiments. Thus, the LTS method is still effective for three-dimensional crack propagation problems.

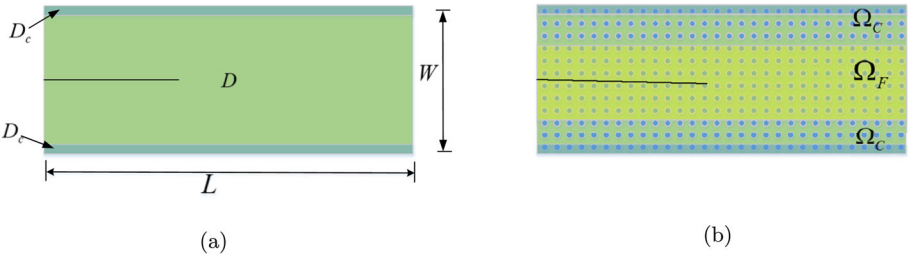


Fig. 16 The geometric configuration of a 2D plate featuring a pre-existing crack

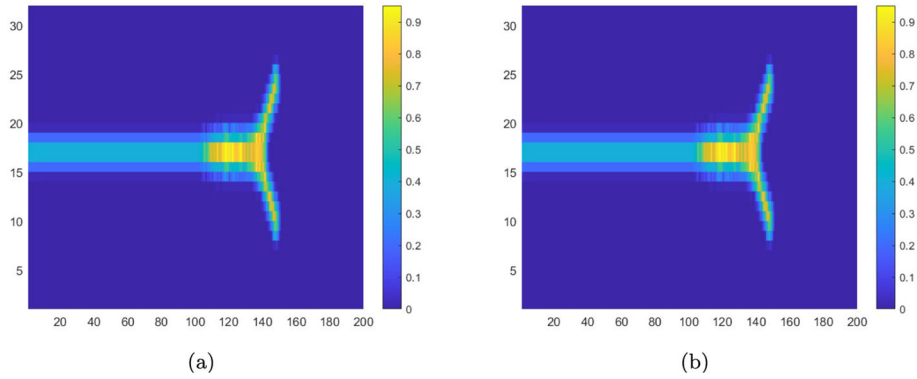


Fig. 17 Crack propagation simulation for OSB-PD with different methods: (a) RK4; (b) LTS4 with $M = 8$

5.7 Simulation of Plate with Pre-Existing Crack using the Ordinary State-Based Peridynamics

We further examine the performance of our LTS methods using a plate with state-based periydnamics. Illustrated in Fig. 16 (a) is the 2D plate under consideration, with dimensions $L = 0.1$ m (length), $W = 0.04$ m (width), $h = 0.01$ m (thickness). A pre-existing crack of length $2c = 0.04$ m is present within the plate. Material properties are $E = 6.5 \times 10^{10}$ Mpa (elastic modulus), density $\rho = 2235$ kg/ m^3 (density), and $\mu = 1/5$ (Poisson's ratio). On the boundary layer D_C, the external loading $b_p = 200$ Mpa is applied, and the width of D_c is 0.001m.

To implement the LTS method, we decompose the region $\Omega = [-L/2, L/2] \times [-W/2, W/2]$ into two distinct subdomains: $\Omega_F = [-L/2, L/2] \times [-W/5, W/5]$ and $\Omega_C = [-L/2, L/2] \times [-W/2, -W/5] \cup [-L/2, L/2] \times [W/5, W/2]$, as shown in Fig. 12 (b). In Ω_F, characterized as a discontinuous region, we employ fine time steps with $\Delta t_f = 1 \times 10^{-8} s$ and spatial grids with $\Delta x_f = 2.5 \times 10^{-4}$ m. On the other hand, Ω_C is identified as a smooth region, governed by coarse time steps with $\Delta t_c = M \Delta t_f$ and $\Delta x_C = M \Delta x_f$. Fig. 17 verifies the accuracy of the proposed algorithm on the state-based model. Here we choose the global time-stepping method, ie., RK4, and LTS4 with $M = 8$ for comparison. It can be observed that the crack shapes obtained by the two methods are consistent, indicating that the proposed method maintains effectiveness on OSB-PD.

Following this, our objective is to assess the inherent computational efficiency advantages of the LTS method compared to the RK method. Keeping the physical time $T = 2 \times 10^{-5} s$, fine time step size $\Delta t_f = 1 \times 10^{-8} s$, and fine grid spacing $\Delta x_f = 2.5 \times 10^{-4}$ m fixed,

Table 6 CPU times for solving the plate with state-based peridynamics, using LTS2, LTS3, and LTS4 with DH-PD

N	M	LTS2	LTS3	LTS4
2000	1	1h20m	1h52m	2h14m
1000	2	53m51s	1h4m33s	1h36m4s
500	4	39m4s	45m29s	1h03m
250	8	22m17s s	36m56s	47m38s

we using various values of $M \in \{1, 2, 4, 8\}$ and compared and analyzed the CPU time. It's important to note that when $M = 1$, the proposed LTS method and the RK method are consistent. Detailed data can be found in Table 6, where N represents the subintervals in the subdomain Ω_C. It's worth noting that the RK method necessitates more time steps to achieve results consistent with the LTS method.

Through the above analysis, we have showcased the accuracy and computational advantages of applying the new algorithm to PD models. By comparing the results of using the meshfree method for spatial discretization and the RK method for temporal discretization in three examples, we have highlighted the effectiveness of our algorithm, particularly in handling discontinuous problems. The proposed algorithm has been demonstrated to be advantageous for simulating discontinuous multi-scale issues while maintaining high accuracy and improving computational efficiency.

6 Conclusions

The primary contribution of this paper lies in the introduction of an efficient algorithm featuring multiple time resolutions for addressing multi-scale crack simulation challenges. This algorithm effectively mitigates the computational burden arising from local refinement. Specifically designed as a local time step method tailored for dual horizon models, it enables the partitioning of regions into distinct sub-regions, allowing for the adoption of varying time steps for solving. Consequently, this approach circumvents the adverse effects of smaller horizons within the region of interest on the minimum time step size. As a result, it significantly enhances computational efficiency while upholding accuracy levels.

The introduction of various examples, including 2D and 3D bond-based models, serves to verify the convergence and computational efficiency of the algorithm. This validates the algorithm's universality, as it is not constrained by specific model structures or spatial sizes. This indicates that the algorithm can demonstrate strong adaptability in addressing multi-scale problems. It should be noted that the proposed method is a temporal method and is independent of spatial discretization methods. Efforts are underway to extend the application of the LTS method to investigate the NOSB-PD model and failure behaviors of composite material.

Appendix A. Convergence Analysis of Second-Order LTS Methods

Proof Let $\mathbf{U}(\mathbf{x}, t)$ represent the exact solution of Eq. (20), and $\hat{\mathbf{U}}_i(t_n)$ denote the numerical solution obtained at t_n, Since $\hat{\mathbf{U}}_i(t_1)$ is defined as obtained by RK methods, we have $\hat{\mathbf{U}}_i(t_1) = \mathbf{U}(\mathbf{x}, t_1)$.

i) Assume $\hat{\mathbf{U}}_i(t_n) = \mathbf{U}(\mathbf{x}, t_n)$, we can establish the local error for $\mathbf{x}_i \in \Omega_C$. To begin, we can express the derivatives of $\hat{\mathbf{U}}_i(t_n)$ as follows:

$$\hat{\mathbf{U}}_i^{'}(t_n) = \mathcal{L}_i\left(\mathbf{Q}_C^n\right), \hat{\mathbf{U}}_i^{''}(t_n) = \mathcal{L}_i^{'}(\mathbf{Q}_C^n)\mathcal{L}_i\left(\mathbf{Q}_C^n\right),$$

Since $\mathcal{L}$ are C^5, we can use Taylor expansion on $\mathcal{L}_i\left(\mathbf{Q}_n^{(1)}\right)$ about t_n and can obtain

$$\begin{aligned}
\mathbf{U}_i^{n+1} =& \mathbf{U}_i^n + \frac{\Delta t}{2}\left(\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \mathcal{L}_i\left(\mathbf{Q}_n^{(1)}\right)\right) \\
=& \mathbf{U}_i^n + \Delta t\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \frac{\Delta t^2}{2}\mathcal{L}_i^{'}(\mathbf{Q}_C^n)\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + O(\Delta t^3) \\
=& \hat{\mathbf{U}}_i(t_n) + \Delta t\hat{\mathbf{U}}_i^{'}(t_n) + \frac{\Delta t^2}{2}\hat{\mathbf{U}}_i^{''}(t_n) + O(\Delta t^3) \\
=& \hat{\mathbf{U}}_i(t_{n+1}) + O(\Delta t^3), \ \mathbf{x}_i \in \Omega_{BC}.
\end{aligned} \tag{61}$$

Therefore, the local error $\mathbf{e}_i^{n+1}$ for $\mathbf{x}_i \in \Omega_{BC}$ is $O(\Delta t^3)$. It's important to note that the values of $\mathbf{x}_i \in \Omega_C$ are obtained using RK2, resulting in a local error within Ω_C of $O(\Delta t^3)$.

ii) Then we verify the local error of the interpolation polynomial. By expanding the Taylor function of parameter β , we have:

$$\beta = \frac{\mathbf{U}_i^{n+1} - \mathbf{U}_i^n - \Delta t\mathcal{L}_i\left(\mathbf{Q}_C^n\right)}{\Delta t^2} = \mathcal{L}_i^{'}(\mathbf{Q}_C^n)\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + O(\Delta t) = \hat{\mathbf{U}}_i^{''}(t_n) + O(\Delta t), \tag{62}$$

Substituting (62) into the expression of $\mathcal{I}_i(t_{n,k})$ and expanding the Taylor function about t_n, we can obtain

$$\begin{aligned}
\mathcal{I}_i(t_{n,k}) =& \mathbf{U}_i^n + k\Delta t_f\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \left(k\Delta t_f\right)^2\beta \\
=& \hat{\mathbf{U}}_i(t_n) + k\Delta t_f\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \frac{\left(k\Delta t_f\right)^2}{2}\mathcal{L}_i^{'}(\mathbf{Q}_C^n)\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + O(\Delta t^3) \\
=& \hat{\mathbf{U}}_i(t_n) + k\Delta t_f\hat{\mathbf{U}}_i^{'}(t_n) + \frac{(k\Delta t_f)^2}{2}\hat{\mathbf{U}}_i^{''}(t_n) + O(\Delta t^3) \\
=& \hat{\mathbf{U}}_i(t_{n,k}) + O(\Delta t^3), \ \mathbf{x}_i \in \Omega_{BC}.
\end{aligned} \tag{63}$$

iii) Then we can establish the local error within the region Ω_F. Analyzing the local error of the derivatives of the interpolating polynomial $\mathcal{I}$ allows us to deduce the following:

$$\mathcal{I}_i^{'}(t_{n,k}) = \hat{\mathbf{U}}_i^{'}(t_{n,k}) + O(\Delta t^2), \ \mathbf{x}_i \in \Omega_{BC}. \tag{64}$$

Substituting (64) to (44a)-(46), the methods for the fine time step size can be rewritten as

i) Stage 1:

$$\overline{\mathbf{U}}_i^{(1,k)} = \begin{cases} \mathbf{U}_i^{n,k} + \Delta t_f\mathcal{L}_i\left(\mathbf{Q}_F^{n,k}\right), & \mathbf{x}_i \in \Omega_{IF}, \\ \mathbf{U}_i^{n,k} + \Delta t_f\hat{\mathbf{U}}_i^{'}(t_{n,k}) + O(\Delta t^3), & \mathbf{x}_i \in \Omega_{BF}, \end{cases} \tag{65}$$

ii) Stage 3:

$$\begin{aligned}\mathbf{U}_i^{n,k+1} =& \mathbf{U}_i^{n,k} + \frac{\Delta t_f}{2}\left(\mathcal{L}_i\left(\mathbf{Q}_F^{n,k}\right) + \mathcal{L}_i\left(\mathbf{Q}_F^{(1,k)}\right)\right) = \hat{\mathbf{U}}_i(t_{n,k+1}) + O(\Delta t^3), \\ & \mathbf{x}_i \in \Omega_F.\end{aligned} \tag{66}$$

Then the local error of material points on the region Ω_F is $O(\Delta t^3)$.

iv) The global error of $\mathbf{x}_i\Omega$ can be determined by considering the local error. Let $\mathbf{E}^{n+1}$ denote the value at t_{n+1} of the local error committed at $t_{n,k}$ and propagated forward by the numerical method. Then, $\mathbf{e}_i^{n+1}$ can be expressed as the sum of $\mathbf{E}_i^{n,k}$ for $k = 1, 2, \ldots, M$. We can derive the following estimation for the accumulated error:

$$\left|\mathbf{e}_i^{n+1}\right| \leqslant \sum_{k=1}^{M}\left|\mathbf{E}_i^{n,k}\right|. \tag{67}$$

According to Theorem I.10.2 in [41], we can obtain

$$\left|\mathbf{E}_i^{n,k}\right| \leqslant \mathbf{e}_i^{B\left(t_{n+1}-t_{n,k}\right)}\left|\mathbf{e}_i^{n,k}\right| \leqslant \mathbf{e}_i^{B\Delta t}\left|\mathbf{e}_i^{n,k}\right|, \tag{68}$$

where B represents an upper bound on the Jacobian of (19). Since $|\mathbf{e}_i^{n,k}| = O(\Delta t^4)$, we can express it as $|\mathbf{e}_i^{n,k}| \leq C_2\Delta t^4$, where C_2 is a constant. Utilizing this relation, we can reformulate the estimate as follows:

$$\left|\mathbf{e}_i^{n+1}\right| \leqslant M\mathbf{e}_i^{B\Delta t}\left|\mathbf{e}_i^{n,k}\right| \leqslant M\left(1 + \Delta t B + \frac{(\Delta t B)^2}{2} + \cdots\right)C_2\Delta t^3 \leqslant C_3\Delta t^3. \tag{69}$$

Here C_3 is a constant. Combining this with Theorem II.3.4 in [41], we obtain the expression for the global error $\mathbf{E}_i$:

$$|\mathbf{E}_i| \leqslant \frac{C_2}{M}\left(\mathbf{e}^{B(t_N-t_0)} - 1\right)\Delta t^2. \tag{70}$$

Then the theorem is proved.

□

Appendix B. Convergence Analysis of Thrid-Order LTS Methods

Proof i) First, we prove that the local error for the material points on the subdomain Ω_C is $O(\Delta t^4)$. Let's consider the derivatives of $\hat{\mathbf{U}}_i(t_n)$:

$$\hat{\mathbf{U}}_i^{'''}(t_n) = \mathcal{L}_i^{''}\left(\mathbf{Q}_C^n\right)\left(\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\right)^2 + \left(\mathcal{L}_i^{'}\left(\mathbf{Q}_C^n\right)\right)^2\mathcal{L}_i\left(\mathbf{Q}_C^n\right),$$

Since $\mathcal{L}$ are C^5, we can use Taylor expansion on $\mathcal{L}_i\left(\mathbf{Q}_n^{(1)}\right)$ and $\mathcal{L}_i\left(\mathbf{Q}_n^{(2)}\right)$ about t_n and can obtain

$$\begin{aligned}
\mathbf{U}_i^{n+1} =&\mathbf{U}_i^n + \frac{\Delta t}{4}\left(\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \frac{3}{2}\mathcal{L}_i\left(\mathbf{Q}_n^{(1)}\right) + \frac{3}{2}\mathcal{L}_i\left(\mathbf{Q}_n^{(2)}\right)\right)\\
=&\mathbf{U}_i^n + \Delta t\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \frac{\Delta t^2}{2}\mathcal{L}_i^{'}(\mathbf{Q}_C^n)\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\\
&+ \frac{\Delta t^3}{6}\left(\mathcal{L}_i^{''}\left(\mathbf{Q}_C^n\right)\left(\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\right)^2 + \left(\mathcal{L}_i^{'}(\mathbf{Q}_C^n)\right)^2\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\right) + O(\Delta t^4) \qquad (71)\\
=&\hat{\mathbf{U}}_i(t_n) + \Delta t\hat{\mathbf{U}}_i^{'}(t_n) + \frac{\Delta t^2}{2}\hat{\mathbf{U}}_i^{''}(t_n) + \frac{\Delta t^3}{6}\hat{\mathbf{U}}_i^{'''}(t_n) + O(\Delta t^4)\\
=&\hat{\mathbf{U}}_i(t_{n+1}) + O(\Delta t^4),\ \ \mathbf{x}_i \in \Omega_{BC}.
\end{aligned}$$

Therefore, the local error $\mathbf{e}_i^{n+1}$ for $\mathbf{x}_i \in \Omega_{BC}$ is $O(\Delta t^4)$. Additionally, since the values of $\mathbf{x}_i \in \Omega_C$ are obtained by RK3, the local error on the subdomain Ω_C is also $O(\Delta t^4)$.

ii) Secondly, by expanding the Taylor function of the parameter γ in (43) of interpolation polynomial $\mathcal{I}_i(t_n)$, we have:

$$\begin{aligned}
\gamma =&\frac{6}{5\Delta t}\left(2\frac{\mathbf{U}_i^{n+1} - \mathbf{U}_i^n - \Delta t\mathcal{L}_i\left(\mathbf{Q}_C^n\right)}{\Delta t^2} - \frac{\mathcal{L}_i\left(\mathbf{Q}_C^n\right) - \mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}\right)\\
=&\frac{6}{5\Delta t}\left(\mathbf{U}_i^{''}(t_n) + \frac{\Delta t}{3}\mathbf{U}_i^{'''}(t_n) - \mathcal{L}_i^{'}(\mathbf{Q}_C^n)\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \right.\\
&\left.\frac{\Delta t^2}{2}\left(\mathcal{L}_i^{''}(\mathbf{Q}_C^n)\left(\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\right)^2 + \left(\mathcal{L}_i^{'}(\mathbf{Q}_C^n)\right)^2\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\right) + O(\Delta t^2)\right)\\
=&\hat{\mathbf{U}}_i^{'''}(t_n) + O(\Delta t),\ \ \mathbf{x}_i \in \Omega_{BC}. \qquad (72)
\end{aligned}$$

Substituting (72) into the expression of $\mathcal{I}_i(t_{n,k})$ and expanding the Taylor function about t_n, we can obtain

$$\begin{aligned}
\mathcal{I}_i(t_{n,k}) =&\mathbf{U}_i^n + k\Delta t_f\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \left(k\Delta t_f\right)^2(\beta - \Delta t\gamma) + \left(k\Delta t_f\right)^3\gamma\\
=&\hat{\mathbf{U}}_i(t_n) + k\Delta t_f\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \frac{\left(k\Delta t_f\right)^2}{2}\mathcal{L}_i^{'}(\mathbf{Q}_C^n)\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \hat{\mathbf{U}}_i^{'''}(t_n) + O(\Delta t^2)\\
=&\hat{\mathbf{U}}_i(t_n) + k\Delta t_f\hat{\mathbf{U}}_i^{'}(t_n) + \frac{(k\Delta t_f)^2}{2}\hat{\mathbf{U}}_i^{''}(t_n) + \frac{(k\Delta t_f)^3}{6}\hat{\mathbf{U}}_i^{'''}(t_n) + O(\Delta t^4)\\
=&\hat{\mathbf{U}}_i(t_{n,k}) + O(\Delta t^4)),\ \ \mathbf{x}_i \in \Omega_{BC}.
\end{aligned}$$

iii) With the interpolating polynomial $\mathcal{I}$, we can obtain

$$\mathcal{I}_i^{'}(t_{n,k}) = \hat{\mathbf{U}}_i^{'}(t_{n,k}) + O(\Delta t^3),\ \ \mathcal{I}^{''}(t_{n,k}) = \hat{\mathbf{U}}_i^{''}(t_{n,k}) + O(\Delta t^2),\ \ \mathbf{x}_i \in \Omega_{BC}. \qquad (73)$$

Substituting (73) to (44a)-(46), the methods for the fine time step size can be rewritten as

i) Stage 1:

$$\overline{\mathbf{U}}_i^{(1,k)} = \begin{cases} \mathbf{U}_i^{n,k} + \dfrac{2\Delta t_f}{3}\mathcal{L}_i\left(\mathbf{Q}_F^{n,k}\right), & \mathbf{x}_i \in \Omega_{IF},\\ \mathbf{U}_i^{n,k} + \dfrac{2\Delta t_f}{3}\hat{\mathbf{U}}_i^{'}(t_{n,k}) + O(\Delta t^4), & \mathbf{x}_i \in \Omega_{BF}, \end{cases} \qquad (74)$$

ii) Stage 2:

$$\overline{\mathbf{U}}_i^{(2,k)} = \begin{cases} \mathbf{U}_i^{n,k} + \dfrac{2\Delta t_f}{3}\mathcal{L}_i\left(\mathbf{Q}_F^{(1,k)}\right), & \mathbf{x} \in \Omega_{IF}, \\ \mathbf{U}_i^{n,k} + \dfrac{2\Delta t_f}{3}\hat{\mathbf{U}}_i^{'}(t_{n,k}) + \dfrac{4\Delta t^2}{9}\hat{\mathbf{U}}_i^{''}(t_{n,k}) + O(\Delta t^4), & \mathbf{x} \in \Omega_{BF}, \end{cases} \tag{75}$$

iii) Stage 3:

$$\begin{aligned} \mathbf{U}_i^{n,k+1} =& \mathbf{U}_i^{n,k} + \left(\frac{\Delta t_f}{4}\right)\left(\mathcal{L}_i\left(\mathbf{Q}_F^{n,k}\right) + \frac{3}{2}\mathcal{L}_i\left(\mathbf{Q}_F^{(1,k)}\right) + \frac{3}{2}\mathcal{L}_i\left(\mathbf{Q}_{(2,k)}^{n}\right)\right) \\ =& \hat{\mathbf{U}}_i(t_{n,k+1}) + O(\Delta t^4)\ , \mathbf{x}_i \in \Omega_F. \end{aligned} \tag{76}$$

Therefore, the local error on Ω_F is $O(\Delta t^4)$.

iv) Finally, we provide a proof of the global error for the material points on the subdomain Ω. As a third-order method, this implies

$$\left\|\mathbf{e}_i^{n+1}\right\| \leqslant M\mathbf{e}_i^{B\Delta t}\left\|\mathbf{e}_i^{n,k}\right\| \leqslant M\left(1 + \Delta t B + \frac{(\Delta t B)^2}{2} + \cdots\right)\cdot C_2\Delta t^4 \leqslant C_3\Delta t^4. \tag{77}$$

and

$$\|\mathbf{E}\| \leqslant \frac{C_2}{M}\left(\mathbf{e}^{B(t_N - t_0)} - 1\right)\Delta t^3. \tag{78}$$

Then the theorem is proved.

□

Appendix C. Convergence Analysis of Fourth-Order LTS Methods

Proof i) First, let's demonstrate that the local error for the material points on the region Ω_C is $O(\Delta t^5)$. We begin by observing that the fourth derivative of $\hat{\mathbf{U}}_i(t_n)$ can be expressed as:

$$\begin{aligned} \hat{\mathbf{U}}_i^{'}(t_n) =& \mathcal{L}_i\left(\mathbf{Q}_C^n\right), \hat{\mathbf{U}}_i^{''}(t_n) = \mathcal{L}_i^{'}(\mathbf{Q}_C^n)\mathcal{L}_i\left(\mathbf{Q}_C^n\right), \\ \hat{\mathbf{U}}_i^{''''}(t_n) =& \mathcal{L}_i^{'''}\left(\mathbf{Q}_C^n\right)\left(\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\right)^3 + 2\mathcal{L}_i^{''}\left(\mathbf{Q}_C^n\right)\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\mathcal{L}_i^{'}\left(\mathbf{Q}_C^n\right)\mathcal{L}_i\left(\mathbf{Q}_C^n\right) \\ &+ 2\mathcal{L}_i^{'}\left(\mathbf{Q}_C^n\right)\mathcal{L}_i^{''}\left(\mathbf{Q}_C^n\right)\mathcal{L}_i^2\left(\mathbf{Q}_C^n\right) + \left(\mathcal{L}_i^{'}\left(\mathbf{Q}_C^n\right)\right)^3\mathcal{L}_i\left(\mathbf{Q}_C^n\right). \end{aligned}$$

Since $\mathcal{L}$ are C^5, we can use Taylor expansion for fourth-order method and obtain

$$\begin{aligned} \mathbf{U}_i^{n+1} =& \mathbf{U}_i^{n+1} + \frac{\Delta t}{4}\left(\mathcal{L}_i\left(\mathbf{Q}_F^n\right) + \frac{3}{2}\mathcal{L}_i\left(\mathbf{Q}_n^{(1)}\right) + \frac{3}{2}\mathcal{L}_i\left(\mathbf{Q}_n^{(2)}\right)\right) \\ =& \mathbf{U}_n^i + \Delta t\mathcal{L}_i\left(\mathbf{Q}_C^n\right) + \frac{\Delta t^2}{2}\mathcal{L}_i^{'}\left(\mathbf{Q}_C^n\right)\mathcal{L}_i\left(\mathbf{Q}_C^n\right) \\ &+ \frac{\Delta t^3}{6}\left(\mathcal{L}_i^{''}\left(\mathbf{Q}_C^n\right)\left(\mathcal{L}_i^{'}\left(\mathbf{Q}_C^n\right)\right)^2 + \left(\mathcal{L}_i^{'}\left(\mathbf{Q}_C^n\right)\right)^2\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\right. \\ &+ \frac{\Delta t^4}{24}\left(\mathcal{L}_i^{'''}\left(\mathbf{Q}_C^n\right)\left(\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\right)^3 + 2\mathcal{L}_i^{''}\left(\mathbf{Q}_C^n\right)\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\mathcal{L}_i^{'}\left(\mathbf{Q}_C^n\right)\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\right. \\ &\left.\left. + 2\mathcal{L}_i^{'}\left(\mathbf{Q}_C^n\right)\mathcal{L}_i^{''}\left(\mathbf{Q}_C^n\right)\mathcal{L}_i^2\left(\mathbf{Q}_C^n\right) + \left(\mathcal{L}_i^{'}\left(\mathbf{Q}^n|_C\right)\right)^3\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\right) + O(\Delta t^5) \end{aligned}$$

$$
\begin{aligned}
&=\hat{\mathbf{U}}_i^n+\Delta t\hat{\mathbf{U}}_i^{'}(t_n)+\frac{\Delta t^2}{2}\hat{\mathbf{U}}_i^{''}(t_n)+\frac{\Delta t^3}{6}\hat{\mathbf{U}}_i^{'''}(t_n)+\frac{\Delta t^4}{24}\hat{\mathbf{U}}_i^{''''}(t_n)+O(\Delta t^5)\\
&=\hat{\mathbf{U}}_i(t_{n+1})+O(\Delta t^5),\ \ \mathbf{x}_i\in\Omega_{BC}.
\end{aligned}
\tag{79}
$$

Similarly, we can obtain $\mathbf{U}_i^{n+1}=\hat{\mathbf{U}}_i(t_{n+1})+O(\Delta t^4)$ and $\mathbf{U}_i^{n+1}=\hat{\mathbf{U}}_i(t_{n+1})+O(\Delta t^3)$ for third-order and second-order methods, respectively. Thus, the local error $\mathbf{e}_i^{n+1}$ for $\mathbf{x}_i\in\Omega_{BC}$ is $O(\Delta t^5)$. It's worth noting that the value of $\mathbf{x}_i\in\Omega_C$ is obtained by RK4, so the local error is also $O(\Delta t^5)$.

ii) Then we prove that the local error of the interpolation polynomial is also $O(\Delta t^5)$. By Taylor expansion, we have

$$
\begin{aligned}
\theta=&\frac{10}{9\Delta t}\Bigg[\frac{6}{5\Delta t}\left(2\frac{\mathbf{U}_i^{n+1}-\mathbf{U}_i^n-\Delta t\mathcal{L}_i\left(\mathbf{U}^n|_{\overline{\Omega}_C}\right)}{\Delta t^2}-\frac{\mathcal{L}_i\left(\mathbf{U}^n|_{\overline{\Omega}_C}\right)-\mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}\right)\\
&-\frac{1}{\Delta t}\left(\frac{\mathcal{L}_i\left(\mathbf{Q}_C^{n}\right)-\mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}-\frac{\mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)-\mathcal{L}_i\left(\mathbf{Q}_C^{n-2}\right)}{\Delta t}\right)\Bigg]\\
=&\mathcal{L}_i^{'''}\left(\mathbf{U}^n|_{\overline{\Omega}_C}\right)\left(\mathcal{L}_i\left(\mathbf{Q}^n|_C\right)\right)^3+2\mathcal{L}_i^{''}\left(\mathbf{U}^n|_{\overline{\Omega}_C}\right)\mathcal{L}_i\left(\mathbf{U}_C^n\right)\mathcal{L}_i^{'}\left(\mathbf{U}^n|_{\overline{\Omega}_C}\right)\mathcal{L}_i\left(\mathbf{Q}_C^n\right)\\
&+2\mathcal{L}_i^{'}\left(\mathbf{U}^n|_{\overline{\Omega}_C}\right)\mathcal{L}_i^{''}\left(\mathbf{U}^n|_{\overline{\Omega}_C}\right)\mathcal{L}_i\left(\mathbf{Q}_C^n\right)+\left(\mathcal{L}_i^{'}\left(\mathbf{Q}_C^n\right)\right)^3\mathcal{L}_i\left(\mathbf{Q}_C^n\right)+O(\Delta t)\\
=&\hat{\mathbf{U}}_i^{''''}(t_n)+O(\Delta t),\\
\epsilon=&\frac{6}{5\Delta t}\left(2\frac{\mathbf{U}_i^{n+1}-\mathbf{U}_i^n-\Delta t\mathcal{L}_i\left(\mathbf{U}^n|_{\overline{\Omega}_C}\right)}{\Delta t^2}-\frac{\mathcal{L}_i\left(\mathbf{U}^n|_{\overline{\Omega}_C}\right)-\mathcal{L}_i\left(\mathbf{Q}_C^{n-1}\right)}{\Delta t}\right)+\frac{11\Delta t}{12}\theta\\
=&\hat{\mathbf{U}}_i^{''}(t_n)+\Delta t\hat{\mathbf{U}}_i^{'''}(t_n)+O(\Delta t^2),\ \ \mathbf{x}_i\in\Omega_{BC}.
\end{aligned}
\tag{80}
$$

Substituting (62), (72) and (80) into the expression of $\mathcal{I}_i(t_{n,k})$ and expanding the Taylor function about t_n, we obtain

$$
\mathcal{I}_i(t_{n,k})=\hat{\mathbf{U}}_i(t_{n,k})+O(\Delta t^5),\ \ \mathbf{x}_i\in\Omega_{BC}.
\tag{81}
$$

Then the local error of the interpolation polynomial is $O(\Delta t^5)$.

iii) Then we can provide a proof for the local error on the region Ω_F. According to (81), we have:

$$
\mathcal{I}_i^{'}(t_{n,k})=\hat{\mathbf{U}}_i^{'}(t_{n,k})+O(\Delta t^4),\ \ \mathcal{I}_i^{''}(t_{n,k})=\hat{\mathbf{U}}_i^{''}(t_{n,k})+O(\Delta t^3),\ \ \mathbf{x}_i\in\Omega_{BC}.
\tag{82}
$$

The for fourth-order method, we have

$$
\overline{\mathbf{U}}_i^{(1,k)}=\begin{cases}\mathbf{U}_i^{n,k}+\dfrac{\Delta t_f}{2}\mathcal{L}_i\left(\mathbf{Q}_F^n\right),\ \ \mathbf{x}_i\in\Omega_F,\\ \overline{\mathbf{U}}_i^{(1,k)}=\dfrac{\Delta t_f}{2}\hat{\mathbf{U}}_i^{'}(t_{n,k})+O(\Delta t^5),,\mathbf{x}_i\in\Omega_{BC},\end{cases}
\tag{83}
$$

$$
\overline{\mathbf{U}}_i^{(2,k)}=\begin{cases}\mathbf{U}_i^{n,k}+\dfrac{\Delta t_f}{2}\mathcal{L}_i\left(\mathbf{Q}_F^{(1,k)}\right),\ \ \mathbf{x}_i\in\Omega_F,\\ \mathbf{U}_i^{n,k}+\dfrac{\Delta t_f}{2}\hat{\mathbf{U}}_i^{'}(t_{n,k})+\dfrac{\Delta t_f^2}{4}\hat{\mathbf{U}}_i^{''}(t_{n,k})+O(\Delta t^5),\ \ \mathbf{x}_i\in\Omega_{BC},\end{cases}
\tag{84}
$$

$$\overline{\mathbf{U}}_i^{(3,k)} = \begin{cases} \mathbf{U}_i^{n,k} + \dfrac{\Delta t_f}{2}\mathcal{L}_i\left(\mathbf{Q}_F^{(2,k)}\right), \ \mathbf{x}_i \in \Omega_F, \\ \mathbf{U}_i^{n,k} + \dfrac{\Delta t_f}{2}\hat{\mathbf{U}}_i'(t_{n,k}) + \dfrac{\Delta t_f^2}{2}\hat{\mathbf{U}}_i''(t_{n,k}) \\ \quad + \dfrac{\Delta t_f^3}{4}\hat{\mathbf{U}}_i'''(t_{n,k}) + O(\Delta t^5), \ \mathbf{x}_i \in \Omega_{BC}, \end{cases} \tag{85}$$

and then can obtain follows by Taylor expansion

$$\mathbf{U}_i^{n,k+1} = \hat{\mathbf{U}}_i(t_{n,k+1}) + O(\Delta t^5), \ , \mathbf{x}_i \in \Omega_F. \tag{86}$$

Then, the local error in Ω_F amounts to $O(\Delta t^5)$.

iv) Finally, we provide a proof for the global error of the material points in the region Ω. We can establish the proof for a third-order method, which implies

$$\left\|\mathbf{e}_i^{n+1}\right\| \leqslant M\mathbf{e}_i^{B\Delta t}\left\|\mathbf{e}_i^{n,k}\right\| \leqslant M\left(1 + \Delta t B + \frac{(\Delta t B)^2}{2} + \cdots\right) \cdot C_2\Delta t^5 \leqslant C_3\Delta t^5. \tag{87}$$

and

$$\|\mathbf{E}\| \leqslant \frac{C_2}{M}\left(\mathbf{e}^{B(t_N - t_0)} - 1\right)\Delta t^4. \tag{88}$$

Then the theorem is proved.

□

Acknowledgements The first author was supported by the the Shandong Provincial Natural Science Foundation (No. ZR2025MS30), the Fundamental Research Funds for the Central Universities (No. 202042008 and No.202264006). The fourth author was supported by the National Natural Science Foundation of China (Grant No 12202051) and the Young Elite Scientists Sponsorship Program by CAST (Grant No 2023QNRC001). The fifth author was as supported by the National Natural Science Foundation of China (No. 12131014), the Natural Science Foundation of Shandong Province (No. ZR2024MA023) and the Fundamental Research Funds for the Central Universities (No. 202264006).

Data Availability The custom codes generated during the current study are available from the corresponding author upon reasonable request. No data sets were generated or analyzed during the current study.

Declarations

Conflict of Interest We declare that we have no financial or personal relationships with others or organizations that can inappropriately influence our work. No professional or other personal interest in any product, service, and/or company could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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